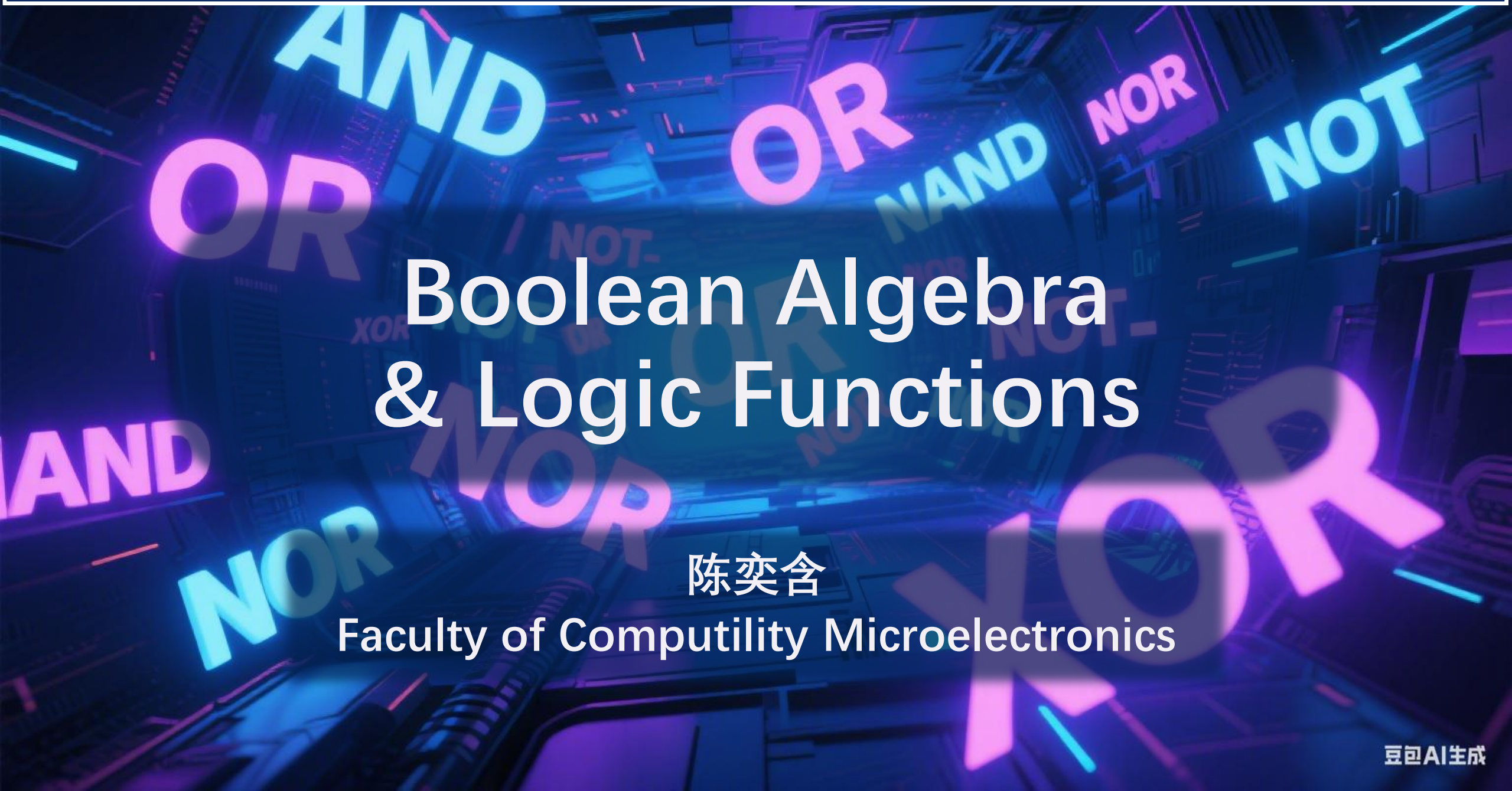


Boolean Algebra & Logic Functions

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Outline

- **Logic Variable**
  - Logic variable, function, operations
- **Representation of Logic Relations**
  - Truth table
  - Logic function
  - Logic gate network
  - Venn diagram
- **Conversion between Logic Representations**
  - Minterm & maxterm
  - Sum-of-product & product-of-sum
  - Cost of logic circuit
- **Boolean Algebra**
  - Axioms, theorems, properties
- **Minimization of Logic Expression**
  - Algebra manipulation
  - Karnaugh map
    - Minimize SOP
    - Minimize POS
    - Incompletely specified functions
- **Multiple-Output Circuits**
- **NAND and NOR Logic Networks**

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Logic Variables

- **Variable and function in arithmetic operation (算术运算)**

$$f(x, y, z) = x^2 + 2y + \sqrt{z}$$

function　　variables: x, y, z are real numbers

- **Logic variable**

Logical variable can be either **True** (“**1**”) or **False** (“**0**”)

$$x = \begin{cases} 1 & True \\ 0 & False \end{cases}$$

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Logic Variables

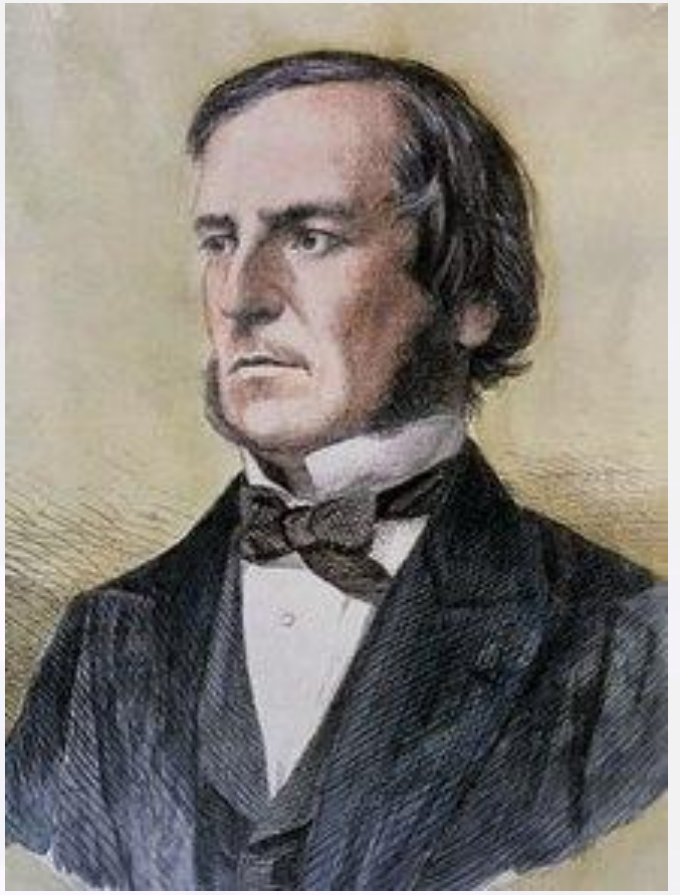
- **Logic variable**

Variables	True (“1”)	False (“0”)
a	Today is Sundy	Today is Monday
b	homework	no homework
c	with friends	without friends
x	happy	unhappy
y	standing	crouching
z	high voltage	low voltage
…	…	…

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Logic Operations

- **Basic logic operation**
  - Boolean algebra (布尔代数): introduced by **George Boole** in his first book
    " The Mathematical Analysis of Logic " (1847)



Operation	Symbol	Rule
NOT 非	$\bar{x}$ $(x' \quad !x \quad \sim x)$	$L(x)=\bar{x}=\begin{cases}1 & if\ x=0\\0 & if\ x=1\end{cases}$
AND 与	$x\cdot y$	$L(x)=x\cdot y=\begin{cases}1 & if\ x=1\ and\ y=1\\0 & otherwise\end{cases}$
OR 或	$x+y$	$L(x)=x+y=\begin{cases}0 & if\ x=0\ and\ y=0\\1 & otherwise\end{cases}$

logic function

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Logic Operations

- **Precedence (优先级) of Operations**
  - In the order of **NOT** > **AND** > **OR**
  - Use parentheses (小括号) if needed
  - Omit the "·" operator when there is no ambiguity

$$\overline{x_1 \cdot \overline{x}_2 + x_3 \cdot x_2 + x_1 \cdot x_4}$$

$$\overline{x_1 \overline{x}_2 + x_3 x_2 + x_1 x_4}$$

$$\overline{x_1 \cdot (\overline{x}_2 + x_3) \cdot x_2 + x_1 \cdot x_4}$$

$$\overline{x_1 (\overline{x}_2 + x_3) x_2 + x_1 x_4}$$

Practice:

Problem: The variables are described as follows.

If *"Today is Sunday"* and I have *"no homework"*, I will be happy.

Or *"Today is Sunday"* and I am *"with friends"*, I will also be happy.

Give the logic function of x in terms of a, b, c.

$$x(a,b,c) = ?$$

Variables	True ("1")	False ("0")
a	Today is Sundy	Today is Monday
b	homework	no homework
c	with friends	without friends
x	happy	unhappy

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Representation of Logic Relations

- **Truth Table (真值表)**

x	$L(x)$
0	1
1	0

x	y	$L(x, y)$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$L(x, y)$
0	0	0
0	1	1
1	0	1
1	1	1

x	y	z	$L(x, y, z)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

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Representation of Logic Relations

- **Logic Function**

x	$L(x)$
0	1
1	0

$$L(x) = \overline{x}$$

x	y	$L(x, y)$
0	0	0
0	1	0
1	0	0
1	1	1

$$L(x, y) = x \cdot y$$

x	y	$L(x, y)$
0	0	0
0	1	1
1	0	1
1	1	1

$$L(x, y) = x + y$$

x	y	z	$L(x, y, z)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

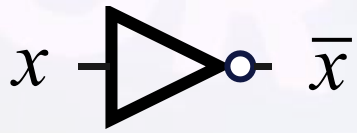
$$L(x, y, z) = x \cdot \overline{y} + x \cdot z$$

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Representation of Logic Relations

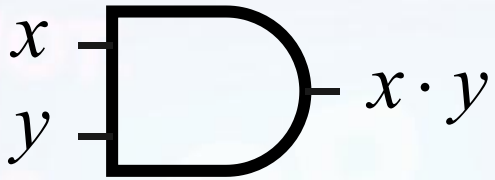
- **Network of Logic Gates**

NOT



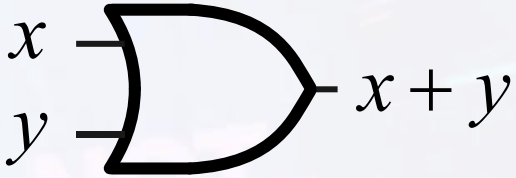
非门/反相器

AND

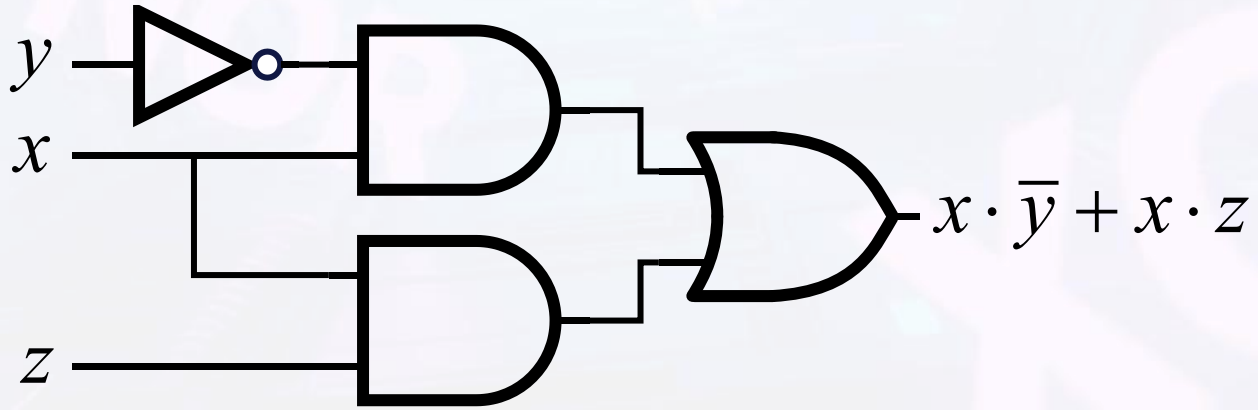


与门

OR



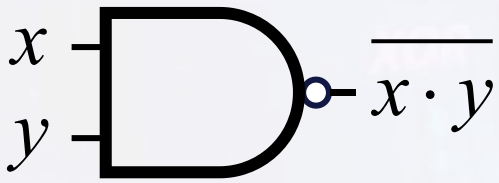
或门



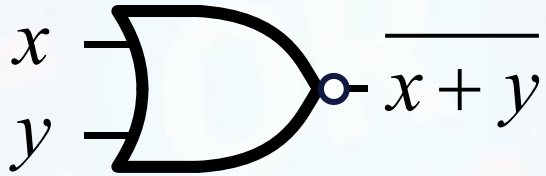
Representation of Logic Relations

- **Network of Logic Gates**
  - More basic logic gates

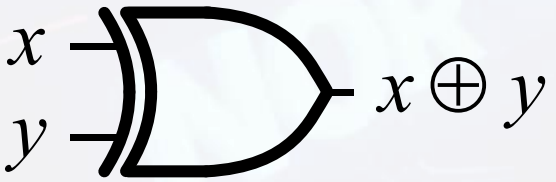
NAND 与非



NOR 或非

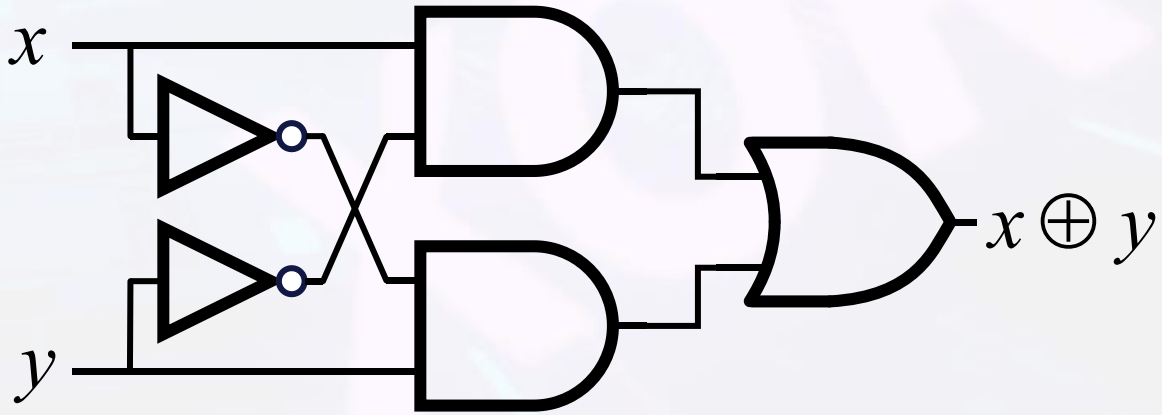


XOR 异或



$$x \oplus y = x \cdot \overline{y} + \overline{x} \cdot y$$

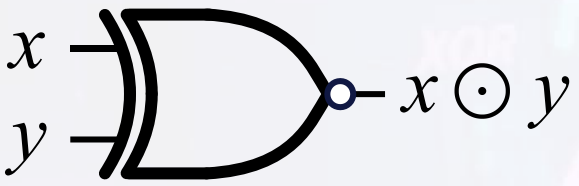
x	y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0



Representation of Logic Relations

- **Network of Logic Gates**
  - More basic logic gates

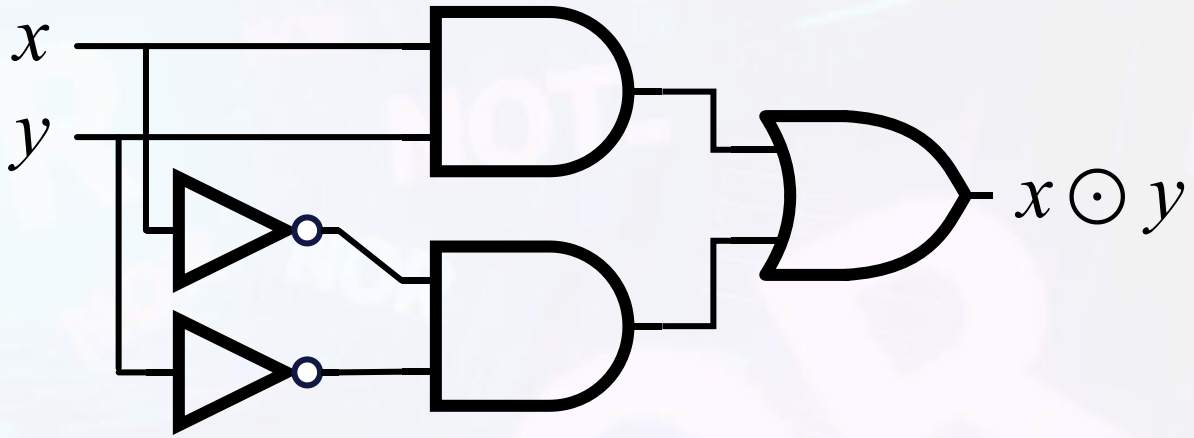
XNOR 同或



$$x \odot y = x \cdot y + \overline{x} \cdot \overline{y}$$

$$= \overline{x \oplus y}$$

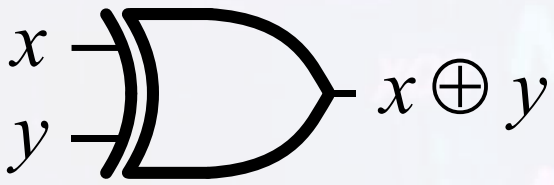
x	y	$x \odot y$
0	0	1
0	1	0
1	0	0
1	1	1



Representation of Logic Relations

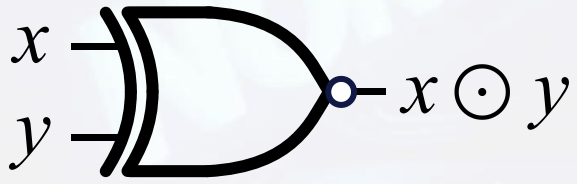
- **Network of Logic Gates**
  - More basic logic gates

XOR 异或



The XOR is sometimes referred to as the **odd function** (奇函数) because the output is equal to 1 if an odd number of inputs have the value 1, and it is equal to 0 otherwise.
输入有奇数个1，输出为1

XNOR 同或



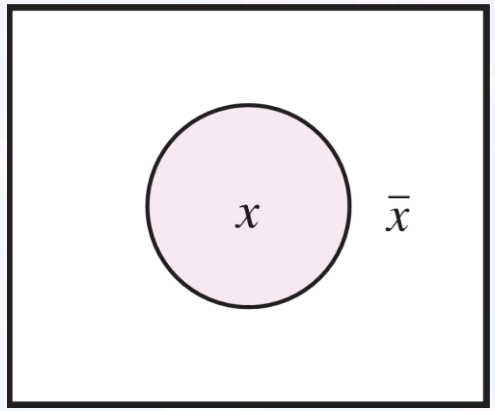
The XNOR is sometimes referred to as the **coincidence operation** (一致性运算) because it produces 1 when its inputs coincide in value.
输入值一致，输出为1

Representation of Logic Relations

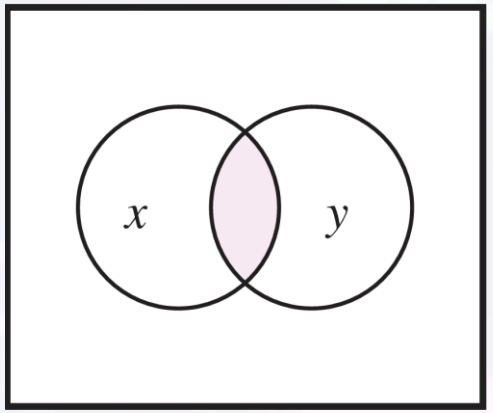
- **Venn Diagram** 韦恩图



1



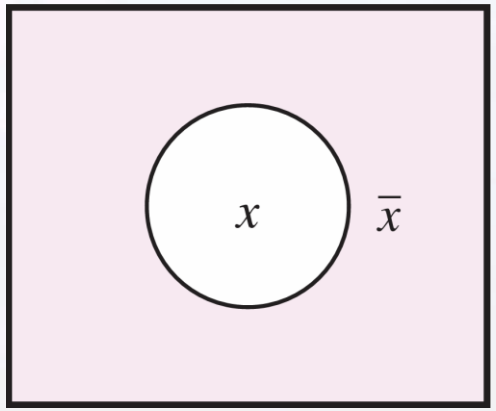
x



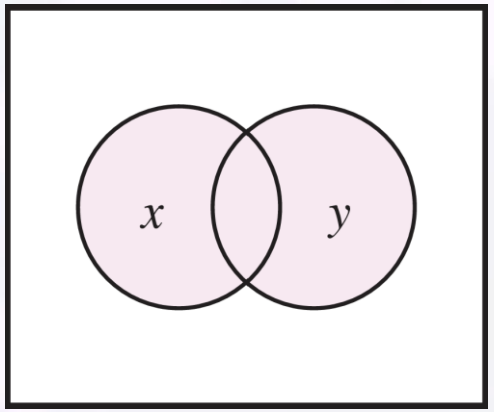
$x \cdot y$



0



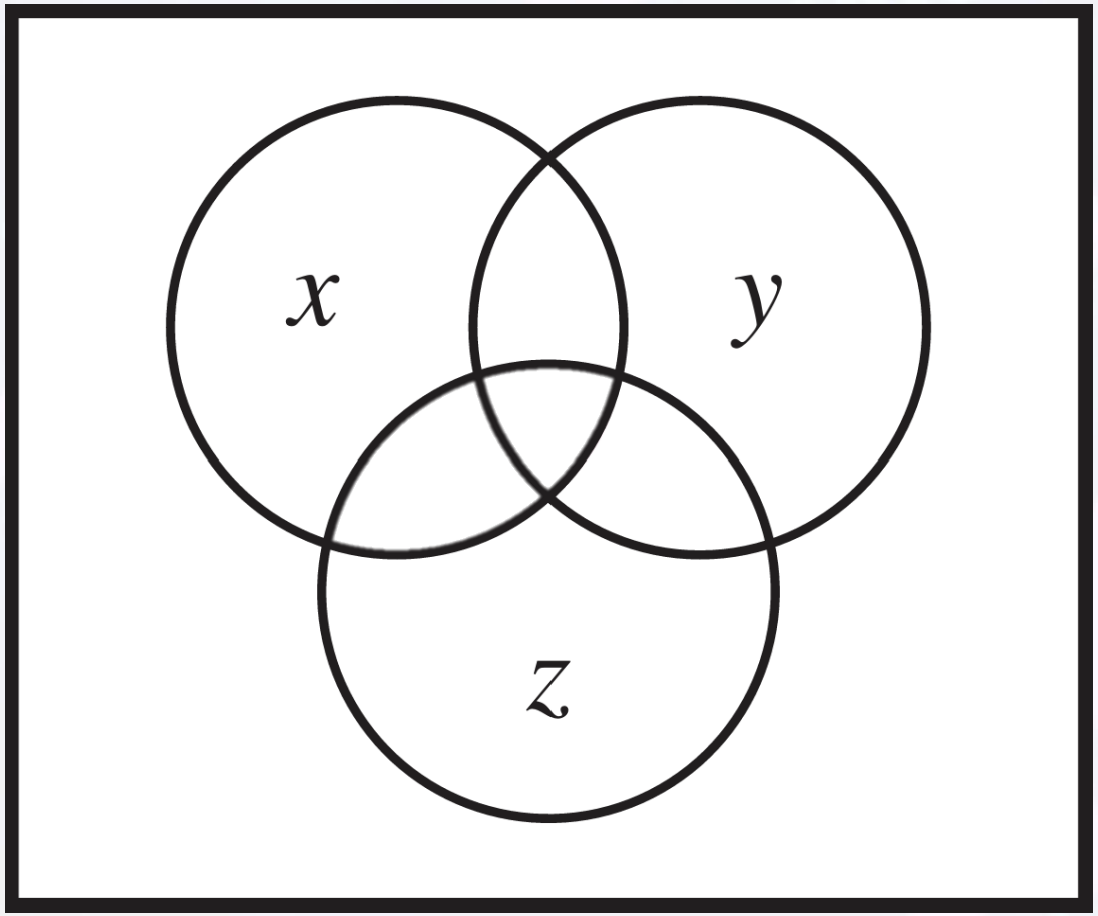
$\overline{x}$



$x + y$

Practice:

Problem: Draw the Venn diagram for $L(x, y, z) = x \cdot \overline{y} + x \cdot z$



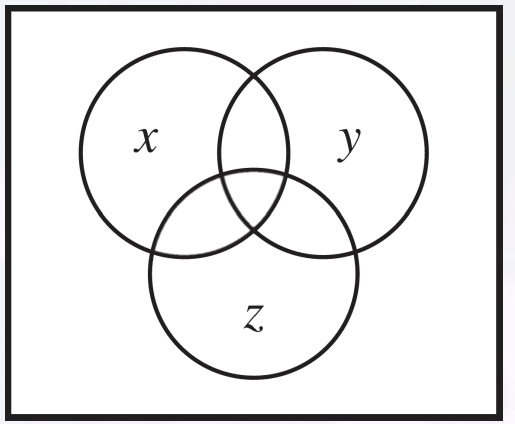
Conversion between Logic Representations

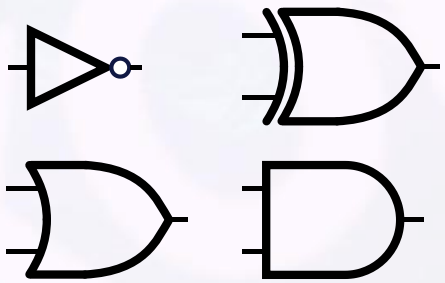
- Logic function as a central form
- Extracting logic function from truth table is not straightforward

x	y	z	$L(x, y, z)$
0	0	0	
0	0	1	…
	…		

?

$L(x, y, z) = \ldots$





The process to generate a circuit that realizes the functional behavior from the description of the behavior is called **Synthesis (综合)**

Conversion between Logic Representations

- **From truth table to logic function**

x	y	$L(x, y)$
0	0	1
0	1	1
1	0	0
1	1	1

$L(x, y) =$ 

x	y	z	$L(x, y, z)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$L(x, y, z) = ?$

Conversion between Logic Representations

- **From truth table to logic function**

w	x	y	z	$L(w, x, y, z)$	w	x	y	z	$L(w, x, y, z)$
0	0	0	0	1	1	0	0	0	1
0	0	0	1	1	1	0	0	1	0
0	0	1	0	1	1	0	1	0	1
0	0	1	1	1	1	0	1	1	0
0	1	0	0	0	1	1	0	0	0
0	1	0	1	0	1	1	0	1	0
0	1	1	0	0	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1

$$L(w, x, y, z) = ?$$

Conversion between Logic Representations

- How to use standard process rather than using eyes

x	y	$L(x, y)$
0	0	1
0	1	1
1	0	0
1	1	1

$\bar{x} \cdot \bar{y} = \bar{0} \cdot \bar{0} = 1$

$\bar{x} \cdot y = \bar{0} \cdot 1 = 1$

$x \cdot y = 1 \cdot 1 = 1$

$L(x, y) = \bar{x} \cdot \bar{y} + \bar{x} \cdot y + x \cdot y$

by analysis

equal

$L(x, y) = \bar{x} + y$

by observation

Boolean Algebra

- **Axioms (公理)**

$$\begin{cases} 1a. & 0 \cdot 0 = 0 \\ 1b. & 1 + 1 = 1 \end{cases}$$

$$\begin{cases} 2a. & 1 \cdot 1 = 1 \\ 2b. & 0 + 0 = 0 \end{cases}$$

$$\begin{cases} 3a. & 0 \cdot 1 = 1 \cdot 0 = 0 \\ 3b. & 1 + 0 = 0 + 1 = 1 \end{cases}$$

$$\begin{cases} 4a. & \text{If } x = 0, \text{ then } \overline{x} = 1 \\ 4b. & \text{If } x = 1, \text{ then } \overline{x} = 0 \end{cases}$$

- **Single-Variable Theorems (定理)**

$$\begin{cases} 5a. & x \cdot 0 = 0 \\ 5b. & x + 1 = 1 \end{cases}$$

$$\begin{cases} 6a. & x \cdot 1 = x \\ 6b. & x + 0 = x \end{cases}$$

$$\begin{cases} 7a. & x \cdot x = x \\ 7b. & x + x = x \end{cases}$$

$$\begin{cases} 8a. & x \cdot \overline{x} = 0 \\ 8b. & x + \overline{x} = 1 \end{cases}$$

$$9. \quad \overline{\overline{x}} = x$$

Boolean Algebra

- **Two-Variable and Three-Variable Properties**

Commutative (交换律)

$$10a.\quad x \cdot y = y \cdot x$$
$$10b.\quad x + y = y + x$$

Associative (结合律)

$$11a.\quad x \cdot (y \cdot z) = (x \cdot y) \cdot z$$
$$11b.\quad x + (y + z) = (x + y) + z$$

Distributive (分配律)

$$12a.\quad x \cdot (y + z) = x \cdot y + x \cdot z$$
$$12b.\quad x + y \cdot z = (x + y) \cdot (x + z)$$

Absorption (吸收律)

$$13a.\quad x + x \cdot y = x$$
$$13b.\quad x \cdot (x + y) = x$$

Combining (组合律)

$$14a.\quad x \cdot y + x \cdot \bar{y} = x$$
$$14b.\quad (x + y) \cdot (x + \bar{y}) = x$$

Boolean Algebra

- **Two-Variable and Three-Variable Properties**

DeMorgan's theorem (德摩根定理、反演律)

$$15a.\quad \overline{x \cdot y} = \bar{x} + \bar{y}$$

$$15b.\quad \overline{x + y} = \bar{x} \cdot \bar{y}$$

$$16a.\quad x + \bar{x} \cdot y = x + y$$

$$16b.\quad x \cdot (\bar{x} + y) = x \cdot y$$

Consensus (包含律)

$$17a.\quad x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z$$

$$17b.\quad (x + y) \cdot (y + z) \cdot (\bar{x} + z) = (x + y) \cdot (\bar{x} + z)$$

Boolean Algebra

- **Principle of Duality (对偶性原理)**
  Given a logic expression, its dual (对偶) is obtained by:
  - replacing all "+" with "·"
  - replacing all "·" with "+"
  - replacing all "0" with "1"
  - replacing all "1" with "0"

  A true statement's dual is also a true statement !

Practice:

Prove the validity of following logic equations:

1. $\overline{x} \cdot \overline{y} + \overline{x} \cdot y + x \cdot y = \overline{x} + y$

2. $x_1 \cdot \overline{x}_3 + \overline{x}_2 \cdot \overline{x}_3 + x_1 \cdot x_3 + \overline{x}_2 \cdot x_3 = \overline{x}_1 \cdot \overline{x}_2 + x_1 \cdot x_2 + x_1 \cdot \overline{x}_2$

3. $x_2 \cdot \overline{x}_3 \cdot x_4 + x_1 \cdot x_3 \cdot x_4 + x_1 \cdot \overline{x}_2 \cdot x_4 = x_2 \cdot \overline{x}_3 \cdot x_4 + x_1 \cdot x_4$

4. $(\overline{x}_1 + x_2 + x_3) \cdot (\overline{x}_1 + \overline{x}_2 + \overline{x}_4) \cdot (\overline{x}_1 + x_3 + x_4)$
   $= (\overline{x}_1 + \overline{x}_2 + \overline{x}_4) \cdot (\overline{x}_1 + x_3)$

Synthesis Process

- **Similarity between logic and arithmetic**

	Arithmetic	*Logic*
·	multiplication (×)	and
+	addition	or
operation	$x \cdot 0 = 0$ $x \cdot 1 = x$	$x + 0 = x$
precedence	$x_1 \cdot x_2 + x_3 \cdot x_2$	
properties	$x \cdot y = y \cdot x$ $x + y = y + x$ $x \cdot (y \cdot z) = (x \cdot y) \cdot z$ $x + (y + z) = (x + y) + z$ $x \cdot (y + z) = x \cdot y + x \cdot z$ $x + (y \cdot z) = (x + y) \cdot (x + z)$	

AND: logical **product** 逻辑积

OR: logical **sum** 逻辑和

Minterms & Maxterms

- **Minterms(最小项) & Maxterms(最大项)**
  - **Minterm**: a product term in which each of the n variables appears
  - **Maxterm**: complements of minterms

Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \bar{x}_1\bar{x}_2\bar{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \bar{x}_1\bar{x}_2x_3$	$M_1 = x_1 + x_2 + \bar{x}_3$
2	0	1	0	$m_2 = \bar{x}_1x_2\bar{x}_3$	$M_2 = x_1 + \bar{x}_2 + x_3$
3	0	1	1	$m_3 = \bar{x}_1x_2x_3$	$M_3 = x_1 + \bar{x}_2 + \bar{x}_3$
4	1	0	0	$m_4 = x_1\bar{x}_2\bar{x}_3$	$M_4 = \bar{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1\bar{x}_2x_3$	$M_5 = \bar{x}_1 + x_2 + \bar{x}_3$
6	1	1	0	$m_6 = x_1x_2\bar{x}_3$	$M_6 = \bar{x}_1 + \bar{x}_2 + x_3$
7	1	1	1	$m_7 = x_1x_2x_3$	$M_7 = \bar{x}_1 + \bar{x}_2 + \bar{x}_3$

$$\bar{m}_i = M_i$$

Sum-of-Products

- **Sum-of-Products (SOP 积之和) Form**
  - A function can be represented by: sum of "minterm **AND** function value"

$$f = \sum m \cdot Value$$

x	y	Minterm	$L(x, y)$
0	0	m_0	1
0	1	m_1	1
1	0	m_2	0
1	1	m_3	1

$$\begin{aligned} L &= m_0 \cdot 1 + m_1 \cdot 1 + m_2 \cdot 0 + m_3 \cdot 1 \\ &= \bar{x} \cdot \bar{y} \cdot 1 + \bar{x} \cdot y \cdot 1 + x \cdot \bar{y} \cdot 0 + x \cdot y \cdot 1 \\ &= \bar{x} \cdot \bar{y} + \bar{x} \cdot y + x \cdot y \\ &= \sum (m_0, m_1, m_3) = \sum m(0,1,3) \end{aligned}$$

Only need to sum up minterms corresponding to the rows with $f = 1$

Practice:

Prove the validity of following logic equations:

1. $\sum m(0,1,3) = \overline{x} + y$
2. $\sum m(1,3,5,6,7) = xy + z$

Product-of-Sums

- **Sum-of-Products (SOP 积之和) Form**
  Sum of minterms corresponding to the rows with $f = 1$

$$L = \sum(m_1, m_3, m_5, m_6, m_7) = \sum m(1,3,5,6,7)$$

- **Product-of-Sums (POS 和之积) Form**
  Product of maxterms corresponding to the rows with $f = 0$

$$L = \prod(M_0, M_2, M_4) = \prod M(0,2,4)$$

Practice: Prove $\sum m(1,3,5,6,7) = \prod M(0,2,4)$

x	y	z	L
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

互动AI生成

Practice:

Prove: $\sum m(1,3,5,6,7) = \prod M(0,2,4)$

x	y	z	L
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$L = \sum(m_1, m_3, m_5, m_6, m_7)$$

$$\bar{L} = \sum(m_0, m_2, m_4)$$

$$L = \bar{\bar{L}} = \overline{\sum(m_0, m_2, m_4)}$$

$$= \overline{m_0 + m_2 + m_4}$$

$$= \bar{m}_0 \cdot \bar{m}_2 \cdot \bar{m}_4$$

$$= M_0 \cdot M_2 \cdot M_4$$

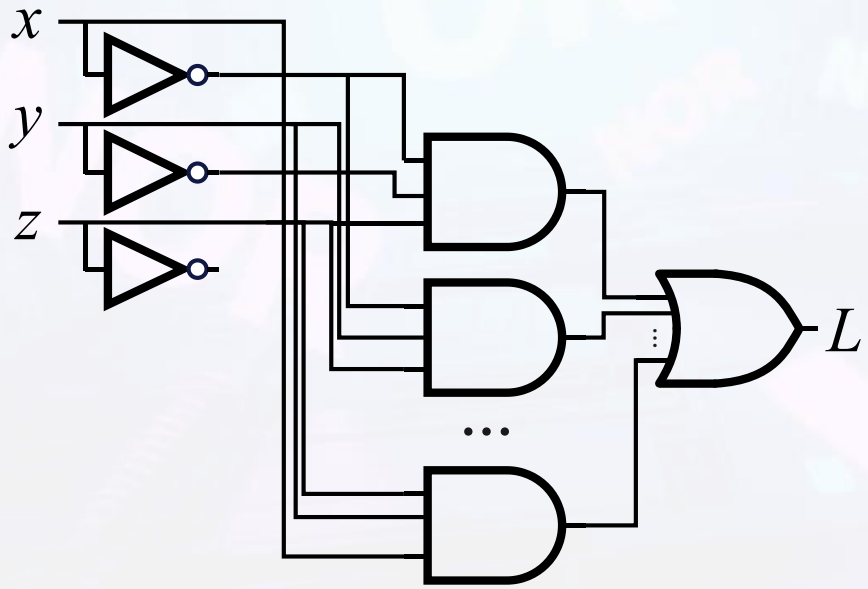
Cost of Logic Circuits

- A good indication of **cost**: the total number of gates plus the total number of inputs to all gates in the circuit

$$\text{Cost} = N_{gates} + N_{inputs}$$

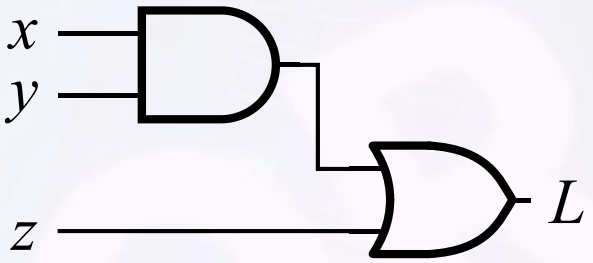
x	y	z	$L(x, y, z)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$L = \sum(m_1, m_3, m_5, m_6, m_7)$$



$$\text{Cost} = 9 + 23 = 32$$

$$L = xy + z$$



$$\text{Cost} = 2 + 4 = 6$$

Cost of Logic Circuits

- How to use standard process rather than using eyes

x	y	$L(x, y)$
0	0	1
0	1	1
1	0	0
1	1	1

by SOP

$$L(x, y) = \overline{x} \cdot \overline{y} + \overline{x} \cdot y + x \cdot y$$

Cost = ?

by observation

$$L(x, y) = \overline{x} + y$$

Cost = ?

Issue: SOP or POS form may not be the one with lowest cost.
Need to be simplified.

Minimization of Logic Expression

- **Karnaugh map (卡诺图)**
  - Using algebraic manipulation to minimize logic function is not obvious, and often tedious and impractical
  - Karnaugh map is a more manageable approach

Two-variable

x_1	x_2	Minterm
0	0	m_0
0	1	m_1
1	0	m_2
1	1	m_3



x_1 \ x_2	0	1
0	m_0	m_1
1	m_2	m_3

Minimization of Logic Expression

- **Karnaugh map (卡诺图)**

Three-variable

x_1	x_2	x_3	Minterm
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7



x_1 \ x_2x_3	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Four-variable

x_1x_2 \ x_3x_4	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

Only one bit change for neighboring cells !

Minimization of Logic Expression

- **Karnaugh map (卡诺图)**

Five-variable

x_1x_2 \ $x_3x_4x_5$	000	001	011	010	110	111	101	100
00	m_0	m_1	m_3	m_2	m_6	m_7	m_5	m_4
01	m_8	m_9	m_{11}	m_{10}	m_{14}	m_{15}	m_{13}	m_{12}
11	m_{24}	m_{25}	m_{27}	m_{26}	m_{30}	m_{31}	m_{29}	m_{28}
10	m_{16}	m_{17}	m_{19}	m_{18}	m_{22}	m_{23}	m_{21}	m_{20}

axis of symmetry (对称轴)

Karnaugh Map

- **Karnaugh map representing logic expression**

$L = x_1\bar{x}_2$

x_1 \ x_2	0	1
0	m_0	m_1
1	m_2	m_3

$L = x_2$

x_1 \ x_2	0	1
0	m_0	m_1
1	m_2	m_3

$L = x_1\bar{x}_2x_3$

x_1 \ x_2x_3	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

$L = x_1\bar{x}_3$

x_1 \ x_2x_3	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

$L = x_2$

x_1 \ x_2x_3	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Karnaugh Map

- **Karnaugh map representing logic expression**

$$L = x_2\bar{x}_3\bar{x}_4$$

x_1x_2 \ x_3x_4	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

$$L = x_1\bar{x}_2$$

x_1x_2 \ x_3x_4	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

$$L = x_1$$

x_1x_2 \ x_3x_4	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

Application of **combining (组合律)** by circling neighboring cells !

Karnaugh Map

- **Karnaugh map representing logic expression**

$$L = \bar{x}_1\bar{x}_2\bar{x}_4$$

x_1x_2 \ x_3x_4	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

$$L = x_2\bar{x}_4$$

x_1x_2 \ x_3x_4	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

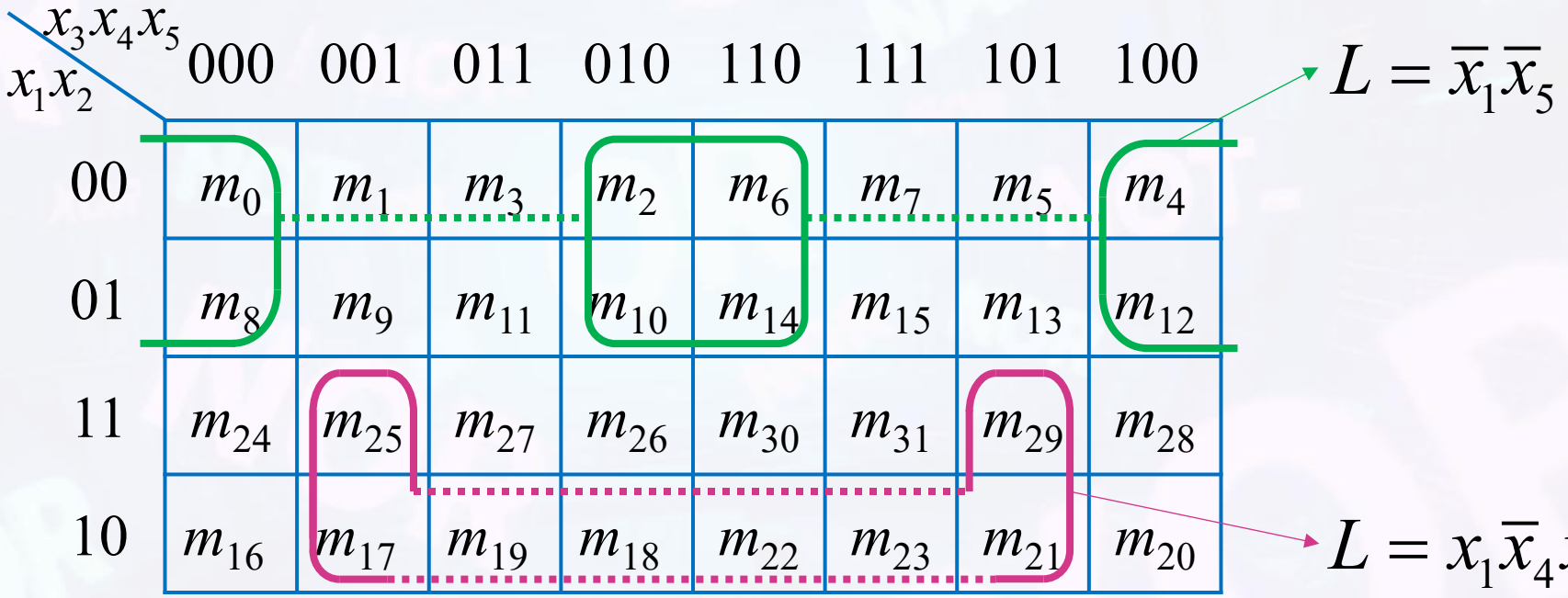
$$L = \bar{x}_2\bar{x}_4$$

x_1x_2 \ x_3x_4	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

Application of **combining (组合律)** by circling neighboring cells !

Karnaugh Map

- **Karnaugh map representing logic expression**



Application of **combining (组合律)** by circling neighboring cells !

Karnaugh Map

- **Minimize SOP with Karnaugh map**

x_1	x_2	x_3		$L(x_1, x_2, x_3)$
0	0	0	m_0	0
0	0	1	m_1	1
0	1	0	m_2	0
0	1	1	m_3	1
1	0	0	m_4	0
1	0	1	m_5	1
1	1	0	m_6	1
1	1	1	m_7	1

Given the truth table:

- **Step1**: mark the minterms with $L=1$

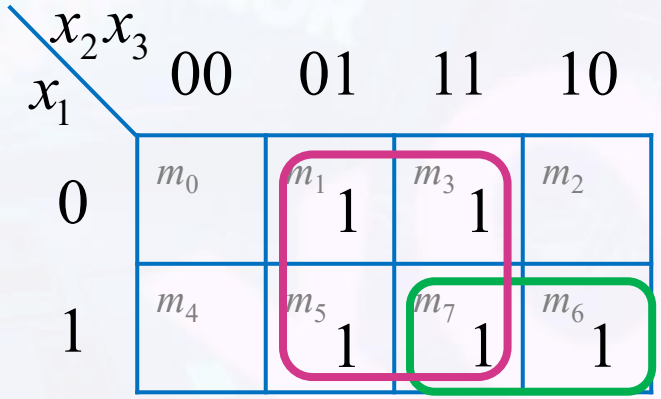
x_1 \ x_2x_3	00	01	11	10
0	m_0	m_1 1	m_3 1	m_2
1	m_4	m_5 1	m_7 1	m_6 1

Karnaugh Map

- **Minimize SOP with Karnaugh map**

x_1	x_2	x_3		$L(x_1, x_2, x_3)$
0	0	0	m_0	0
0	0	1	m_1	1
0	1	0	m_2	0
0	1	1	m_3	1
1	0	0	m_4	0
1	0	1	m_5	1
1	1	0	m_6	1
1	1	1	m_7	1

- **Step2**: circle neighboring minterms
  - Circles as **large** as possible
  - **All "1"** cells need to be circled in
  - **Overlap** is encouraged

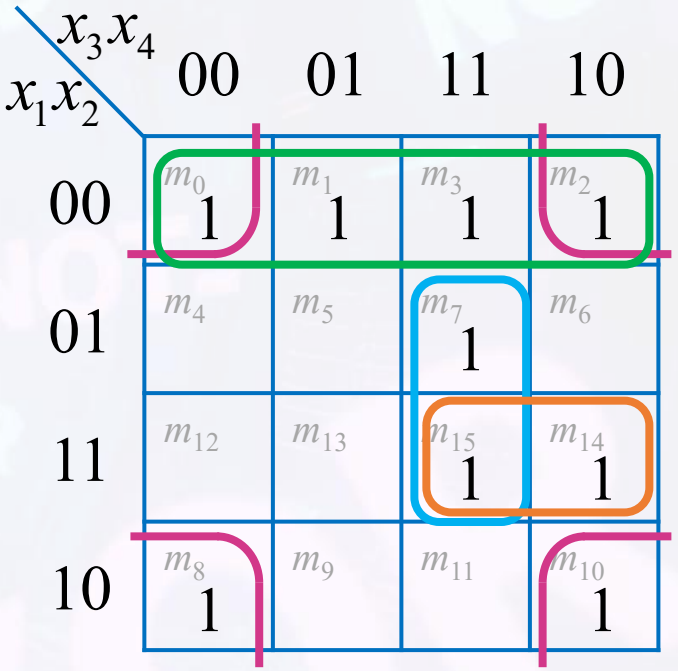


$$L = x_3 + x_1x_2$$

Karnaugh Map

- **Minimize SOP with Karnaugh map**

x_1	x_2	x_3	x_4		L	x_1	x_2	x_3	x_4		L
0	0	0	0	m_0	1	1	0	0	0	m_8	1
0	0	0	1	m_1	1	1	0	0	1	m_9	0
0	0	1	0	m_2	1	1	0	1	0	m_{10}	1
0	0	1	1	m_3	1	1	0	1	1	m_{11}	0
0	1	0	0	m_4	0	1	1	0	0	m_{12}	0
0	1	0	1	m_5	0	1	1	0	1	m_{13}	0
0	1	1	0	m_6	0	1	1	1	0	m_{14}	1
0	1	1	1	m_7	1	1	1	1	1	m_{15}	1



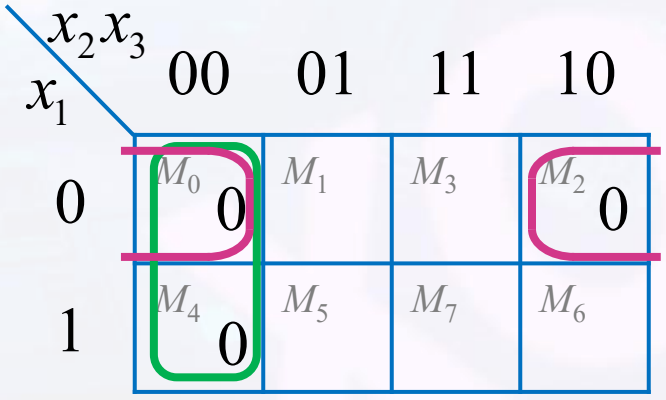
$$L = \overline{x}_2\overline{x}_4 + \overline{x}_1\overline{x}_2 + x_2x_3x_4 + x_1x_2x_3$$

Karnaugh Map

- **Minimize POS with Karnaugh map**

x_1	x_2	x_3		$L(x_1, x_2, x_3)$
0	0	0	M_0	0
0	0	1	M_1	1
0	1	0	M_2	0
0	1	1	M_3	1
1	0	0	M_4	0
1	0	1	M_5	1
1	1	0	M_6	1
1	1	1	M_7	1

- **Step1**: mark the maxterms with $L=0$
- **Step2**: circle neighboring maxterms
  - Circles as **large** as possible
  - **All "0"** cells need to be circled in
  - **Overlap** is allowed

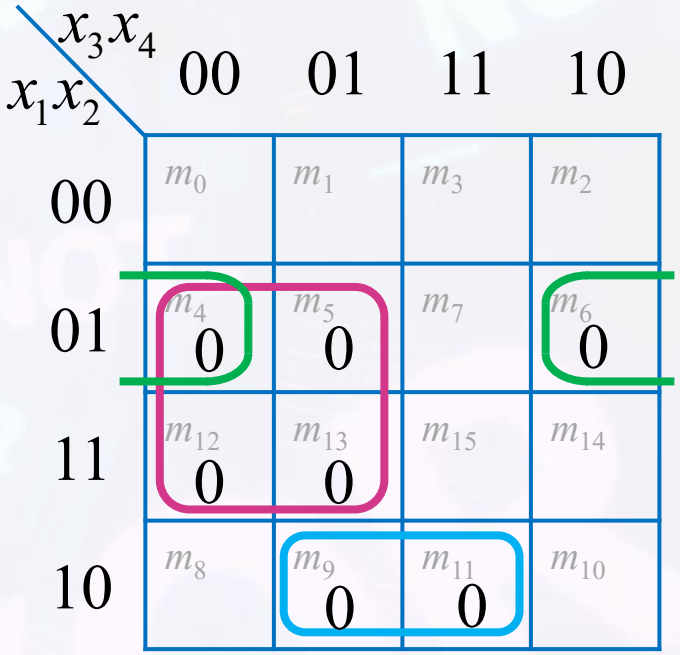


$$L = (x_2 + x_3) \cdot (x_1 + x_3)$$

Karnaugh Map

- **Minimize POS with Karnaugh map**

x_1	x_2	x_3	x_4		L	x_1	x_2	x_3	x_4		L
0	0	0	0	M_0	1	1	0	0	0	M_8	1
0	0	0	1	M_1	1	1	0	0	1	M_9	0
0	0	1	0	M_2	1	1	0	1	0	M_{10}	1
0	0	1	1	M_3	1	1	0	1	1	M_{11}	0
0	1	0	0	M_4	0	1	1	0	0	M_{12}	0
0	1	0	1	M_5	0	1	1	0	1	M_{13}	0
0	1	1	0	M_6	0	1	1	1	0	M_{14}	1
0	1	1	1	M_7	1	1	1	1	1	M_{15}	1



$$L = (\bar{x}_2 + x_3) \cdot (x_1 + \bar{x}_2 + x_4) \cdot (\bar{x}_1 + x_2 + \bar{x}_4)$$

Practice:

Based on the truth table, derive **minimum-cost** expressions for the function: ① in SOP form ② in POS form.

x	y	z	$L(x, y, z)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Practice:

Determine the **minimum-cost** SOP and POS expressions for the function $f(x_1, x_2, x_3, x_4) = \sum m(4, 6, 8, 10, 11, 12, 15)$.

Incompletely Specified Functions

- Sometimes certain input conditions can never occur

  e.g. $(x_1, x_2) = 00,\ 01,\ 10$ $\qquad$ $(x_1, x_2) \neq 11$

- $(x_1, x_2) = 11$ is called a "**don't-care condition**" (无关项), denoted by the letter d in the map

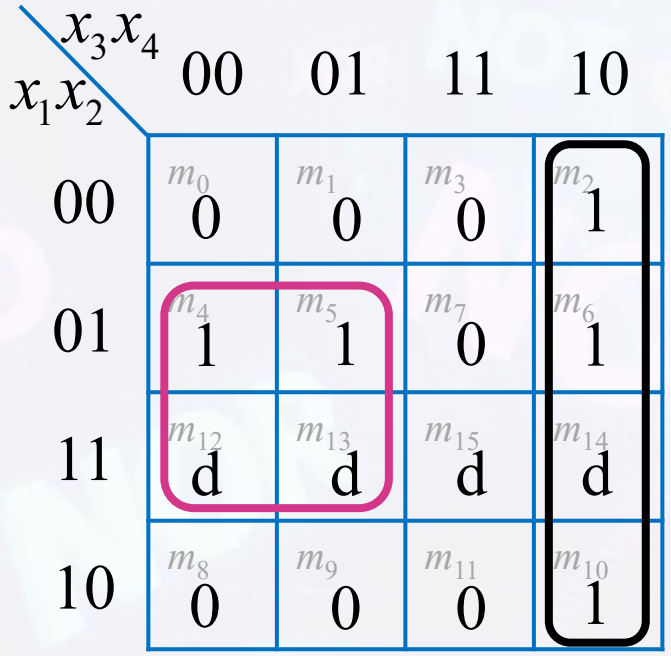
- A function that has don't-care condition(s) is called **incompletely specified function** (非完整定义函数)

- The designer may assume that the function value for these valuations is either 1 or 0

Incompletely Specified Functions

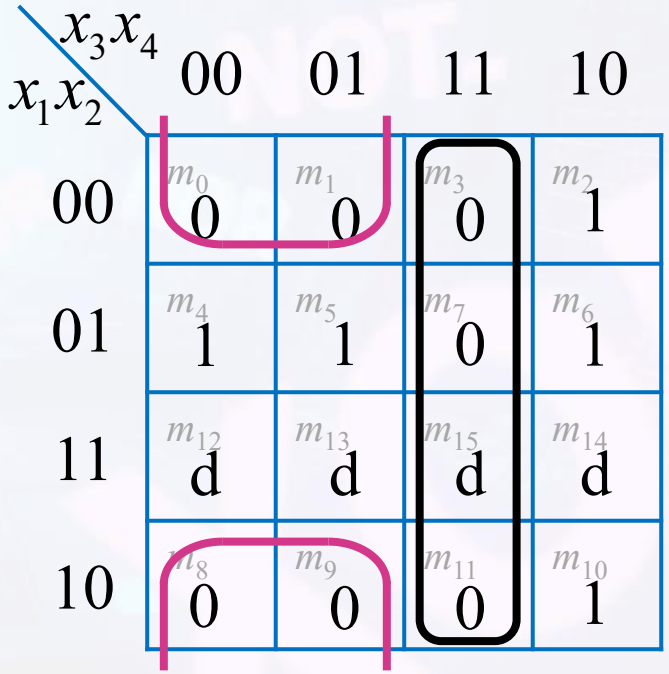
$$L = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$$

SOP implementation



$$L = x_2\bar{x}_3 + x_3\bar{x}_4$$

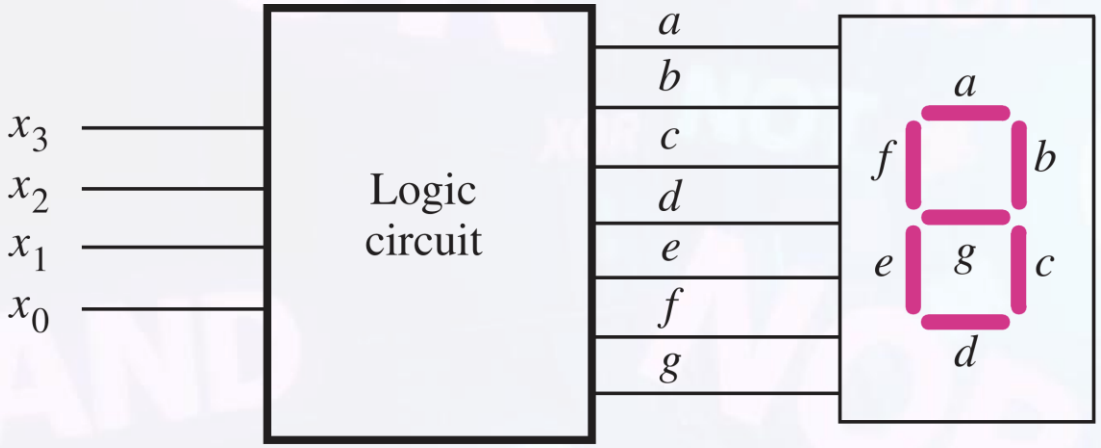
POS implementation



$$L = (x_2 + x_3) \cdot (\bar{x}_3 + \bar{x}_4)$$

Example of Logic Circuit Design

Problem: Design a logic circuit that displays the decimal value of a two-bit number on a seven-segment display.



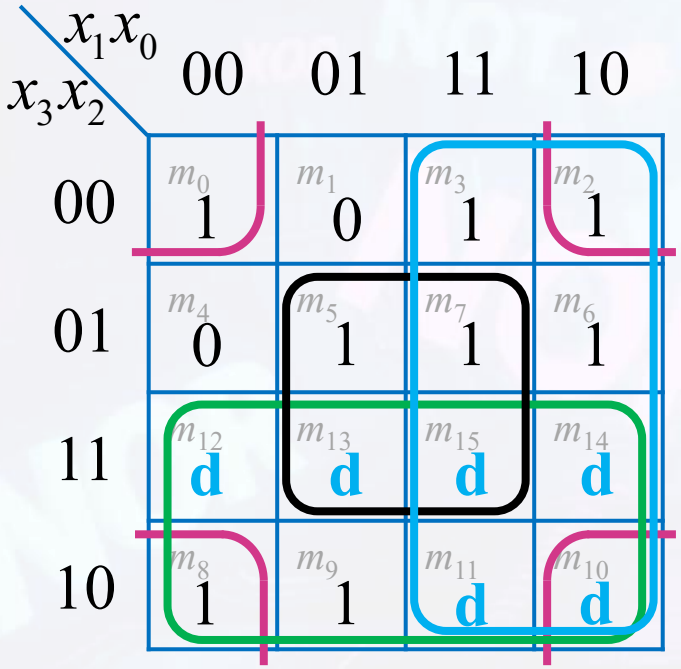
	x_3	x_2	x_1	x_0	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

Example of Logic Circuit Design

Problem: Design a logic circuit that displays the decimal value of a four-bit number on a seven-segment display.

Solution:

For "*a*" segment



$$L = \bar{x}_2\bar{x}_0 + x_2x_0 + x_3 + x_1$$

	x_3	x_2	x_1	x_0	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

Example of Logic Circuit Design

Solution:

For "*b*" segment

x_3x_2 \ x_1x_0	00	01	11	10
00	m_0 1	m_1 1	m_3 1	m_2 1
01	m_4 1	m_5 0	m_7 1	m_6 0
11	m_{12} d	m_{13} d	m_{15} d	m_{14} d
10	m_8 1	m_9 1	m_{11} d	m_{10} d

$L = ?$

For "*c*" segment

x_3x_2 \ x_1x_0	00	01	11	10
00	m_0 1	m_1 1	m_3 1	m_2 0
01	m_4 1	m_5 1	m_7 1	m_6 1
11	m_{12} d	m_{13} d	m_{15} d	m_{14} d
10	m_8 1	m_9 1	m_{11} d	m_{10} d

$L = ?$

For "*d*" segment

x_3x_2 \ x_1x_0	00	01	11	10
00	m_0 1	m_1 0	m_3 1	m_2 1
01	m_4 0	m_5 1	m_7 0	m_6 1
11	m_{12} d	m_{13} d	m_{15} d	m_{14} d
10	m_8 1	m_9 1	m_{11} d	m_{10} d

$L = ?$

Example of Logic Circuit Design

Solution:

For "*e*" segment

x_3x_2 \ x_1x_0	00	01	11	10
00	m_0 1	m_1 0	m_3 0	m_2 1
01	m_4 0	m_5 0	m_7 0	m_6 1
11	m_{12} d	m_{13} d	m_{15} d	m_{14} d
10	m_8 1	m_9 0	m_{11} d	m_{10} d

$L = ?$

For "*f*" segment

x_3x_2 \ x_1x_0	00	01	11	10
00	m_0 1	m_1 0	m_3 0	m_2 0
01	m_4 1	m_5 1	m_7 0	m_6 1
11	m_{12} d	m_{13} d	m_{15} d	m_{14} d
10	m_8 1	m_9 1	m_{11} d	m_{10} d

$L = ?$

For "*g*" segment

x_3x_2 \ x_1x_0	00	01	11	10
00	m_0 0	m_1 0	m_3 1	m_2 1
01	m_4 1	m_5 1	m_7 0	m_6 1
11	m_{12} d	m_{13} d	m_{15} d	m_{14} d
10	m_8 1	m_9 1	m_{11} d	m_{10} d

$L = ?$

互动AI生成

Multiple-Output Circuits

- Possible to **share** some of the gates instead of implementing each function separately

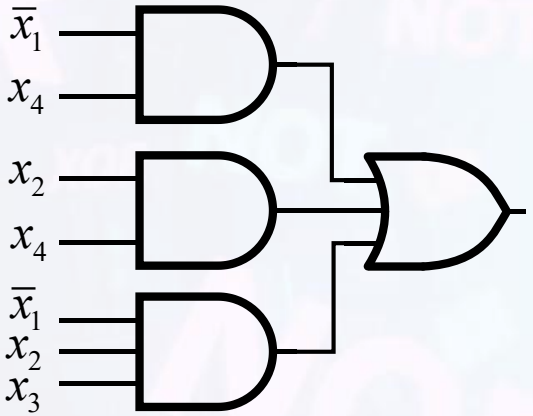
Problem: In a system, two functions f_1 and f_2 are of the same variables, find the minimum-overall-cost implementation.

$$f_1 = \bar{x}_1 x_4 + x_2 x_4 + \bar{x}_1 x_2 x_3$$

$$f_2 = x_1 x_4 + \bar{x}_2 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$$

Multiple-Output Circuits

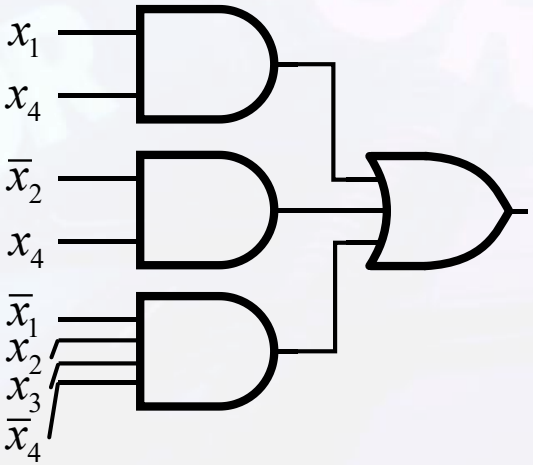
Problem: In a system, two functions f_1 and f_2 are of the same variables, find the minimum-overall-cost implementation.



$$f_1 = \bar{x}_1 x_4 + x_2 x_4 + \bar{x}_1 x_2 x_3$$

Cost = 4 + 10 = 14

Overall Cost = 29

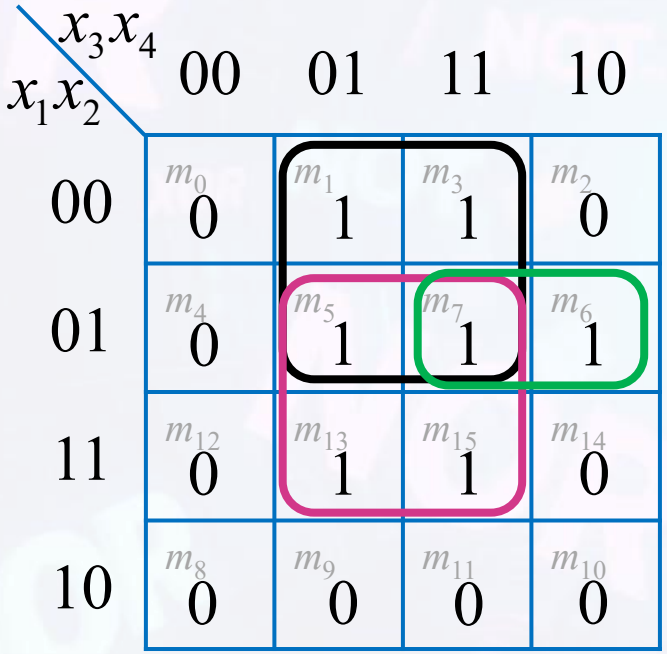


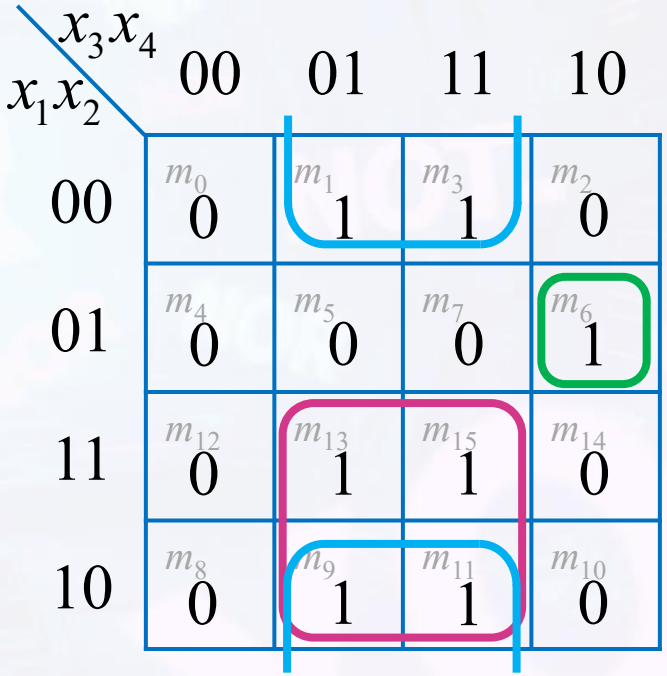
$$f_2 = x_1 x_4 + \bar{x}_2 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$$

Cost = 4 + 11 = 15

Multiple-Output Circuits

Problem: In a system, two functions f_1 and f_2 are of the same variables, find the minimum-overall-cost implementation.



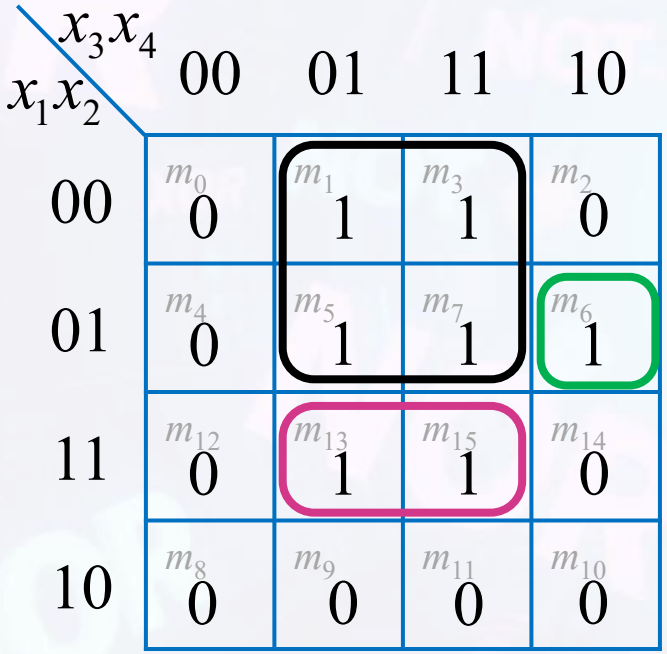


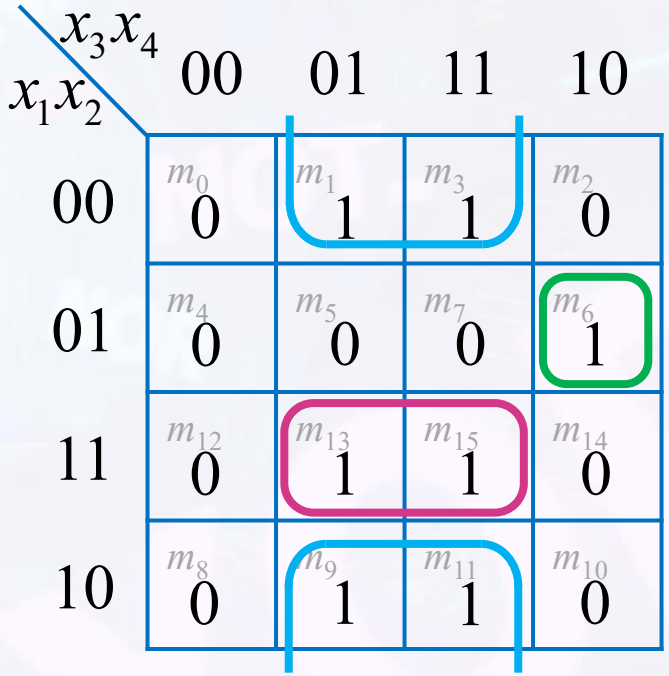
$$f_1 = \bar{x}_1 x_4 + x_2 x_4 + \bar{x}_1 x_2 x_3 \qquad f_2 = x_1 x_4 + \bar{x}_2 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$$

No common gates!

Multiple-Output Circuits

Problem: In a system, two functions f_1 and f_2 are of the same variables, find the minimum-overall-cost implementation.



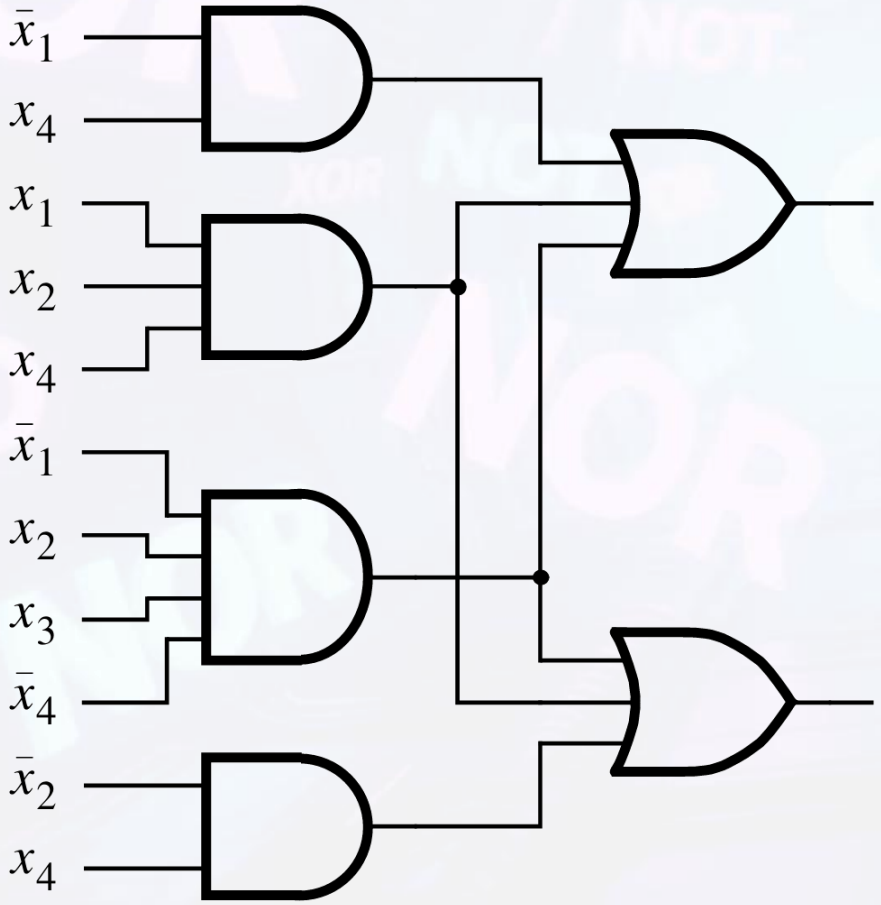


$$f_1 = \bar{x}_1 x_4 + x_1 x_2 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$$

$$f_2 = \bar{x}_2 x_4 + x_1 x_2 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$$

Multiple-Output Circuits

Problem: In a system, two functions f_1 and f_2 are of the same variables, find the minimum-overall-cost implementation.



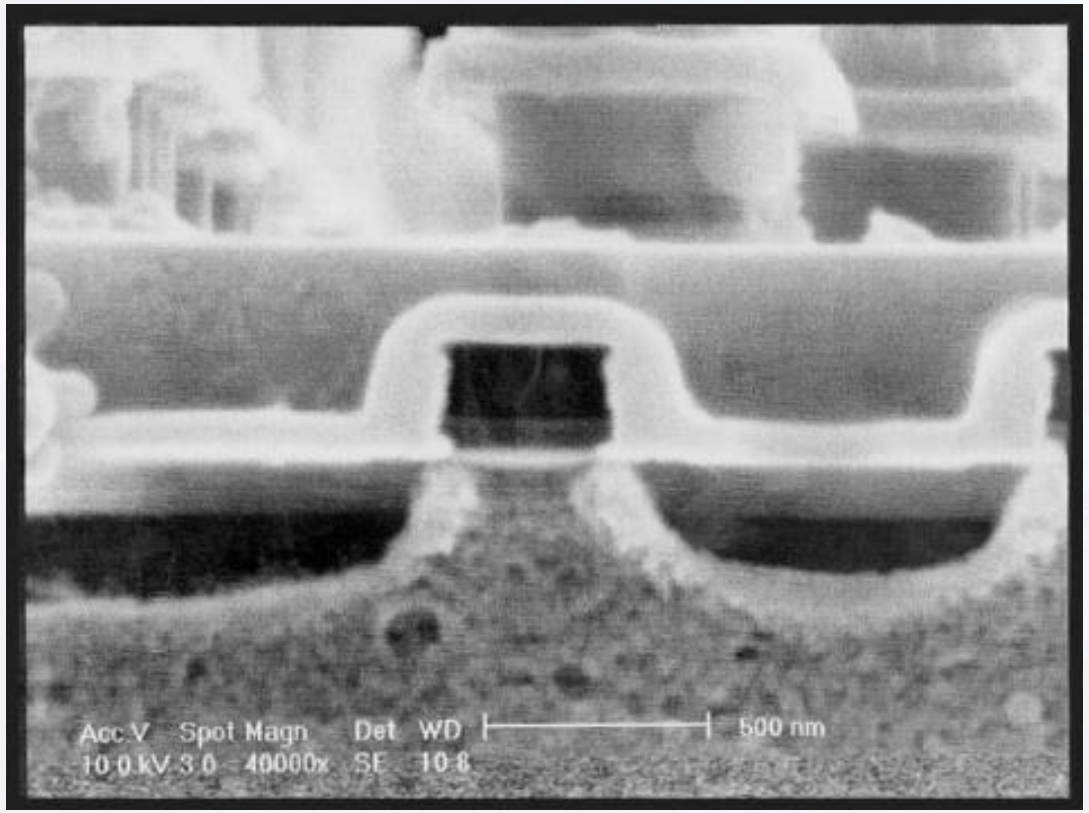
$$f_1 = \bar{x}_1 x_4 + x_1 x_2 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$$

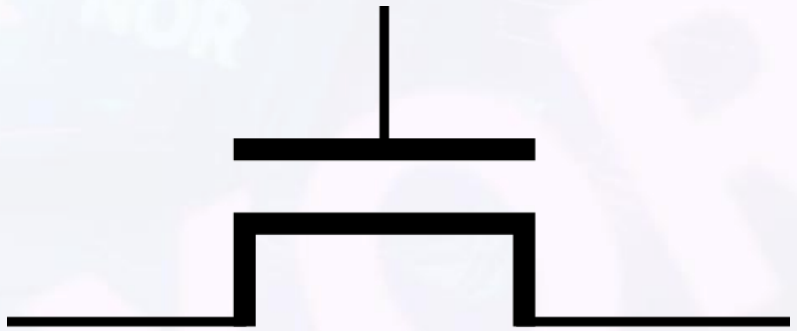
$$f_2 = \bar{x}_2 x_4 + x_1 x_2 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$$

Overall Cost = 6 + 17 = 23 < 29

NAND and NOR Logic Networks

- In practical circuit, **NAND** and **NOR** gates are largely used because they are realized with simpler circuits than AND and OR gates
- Transistors (晶体管) can be seen as switches (开关)

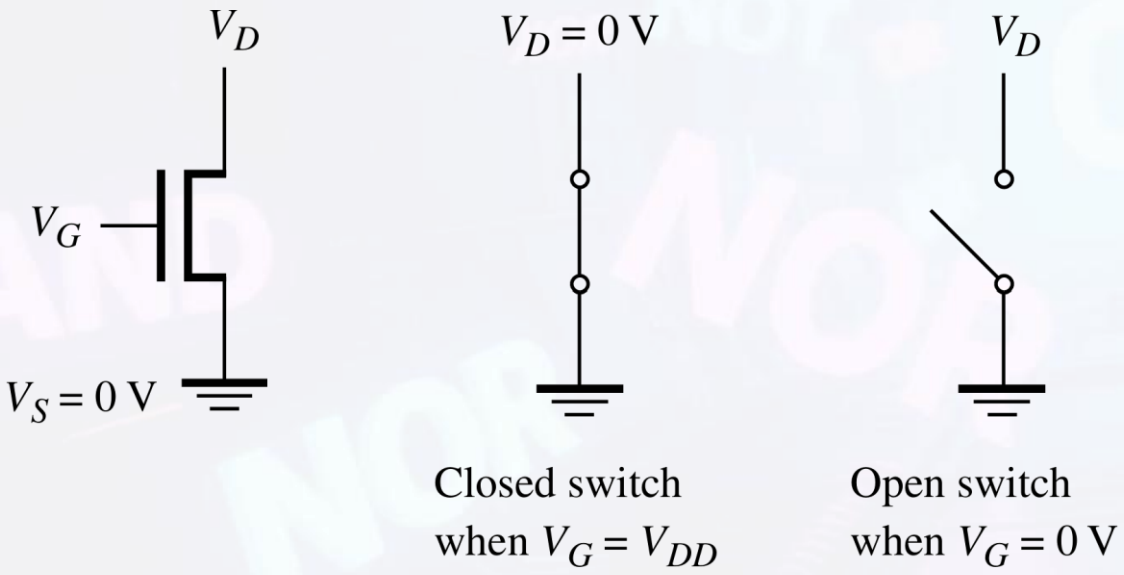




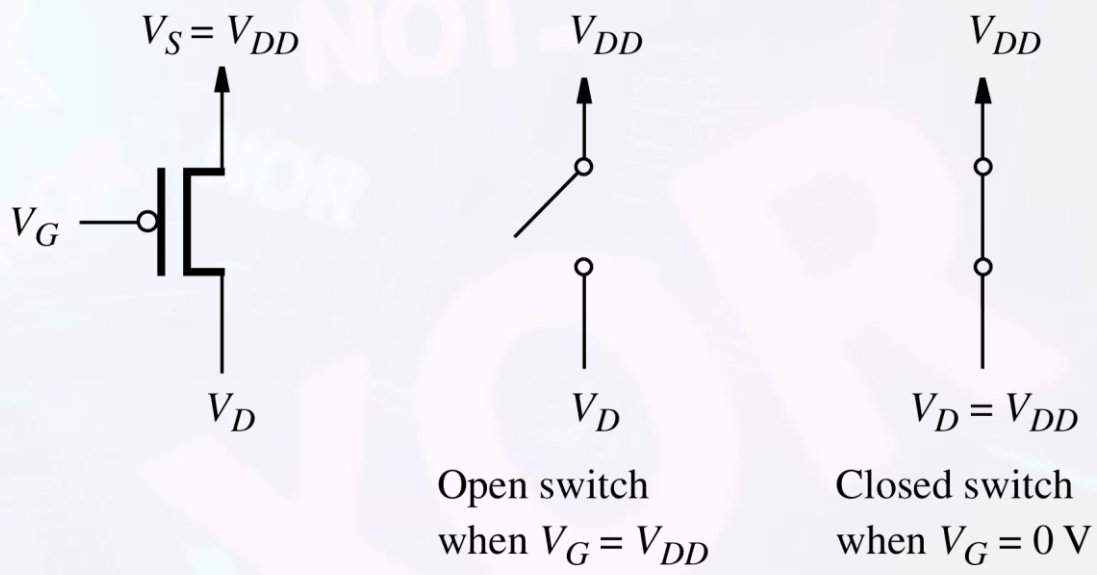
NAND and NOR Logic Networks

- Transistors (晶体管) can be seen as switches (开关)

NMOS transistor

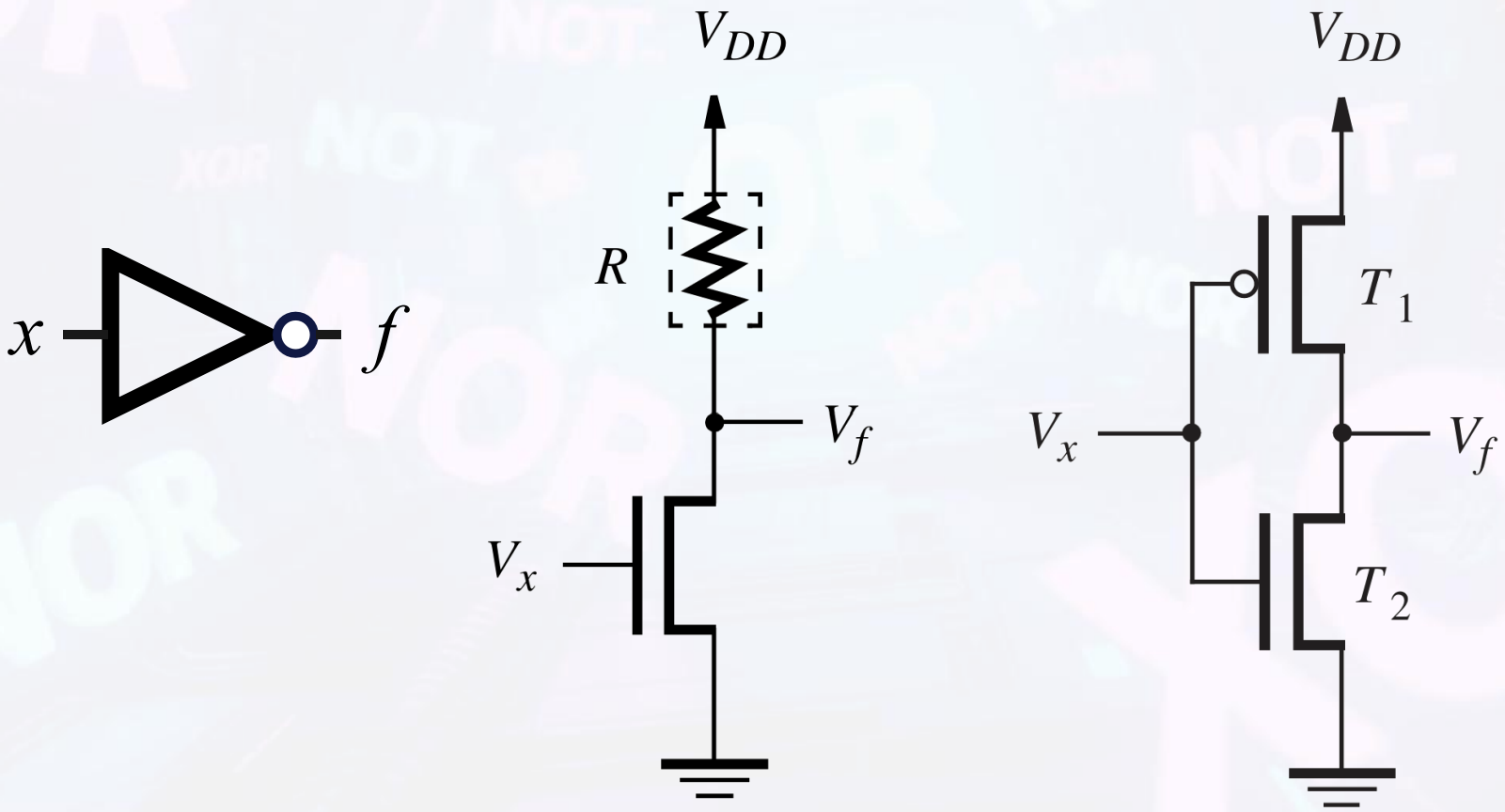


PMOS transistor



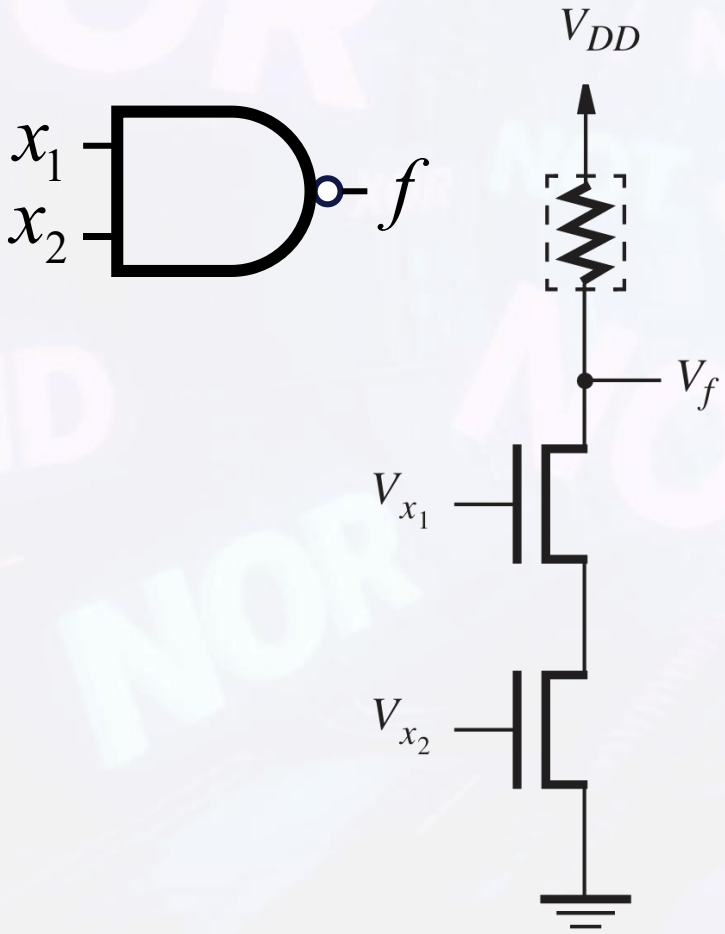
NAND and NOR Logic Networks

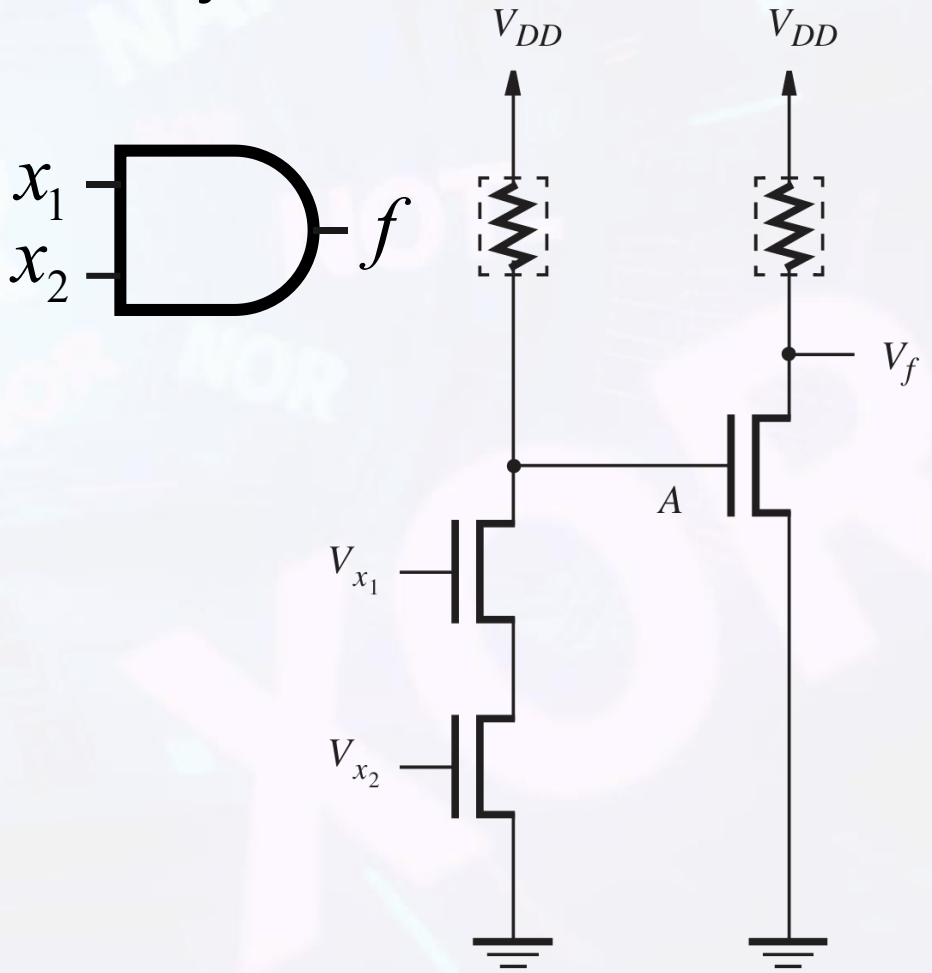
- **Transistors construct logic gates**
  - **NOT** gate with NMOS only or with “NMOS + PMOS”



NAND and NOR Logic Networks

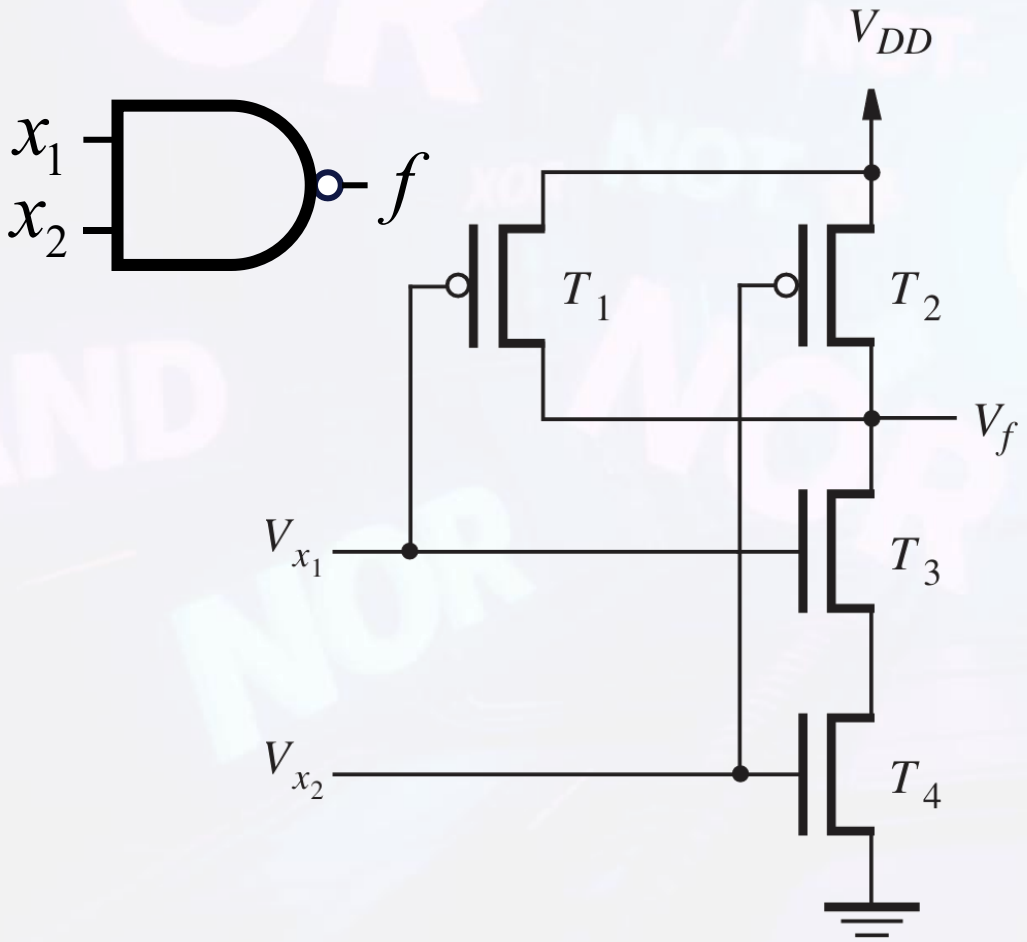
- **Transistors construct logic gates**
  - **NAND** and **AND** gates with NMOS only

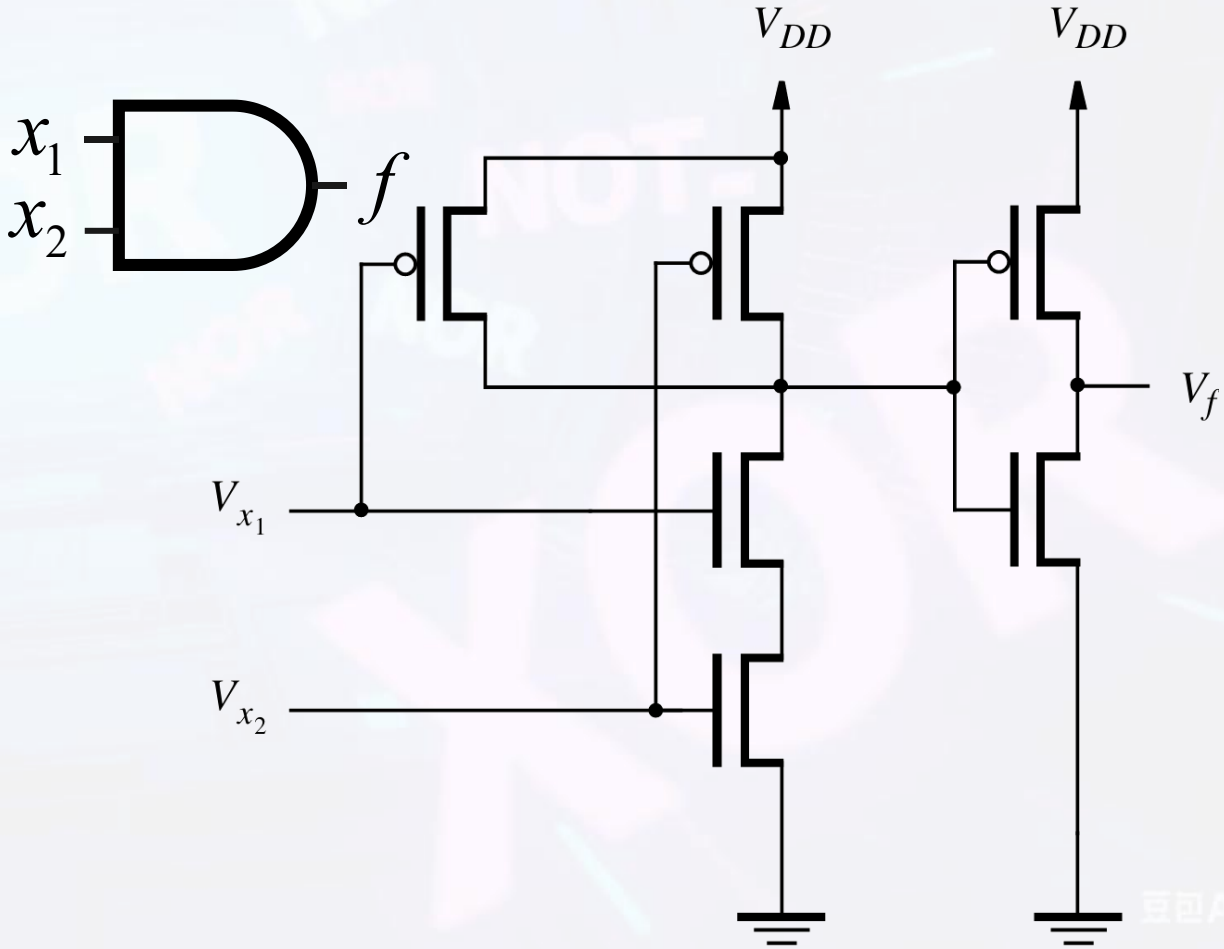




NAND and NOR Logic Networks

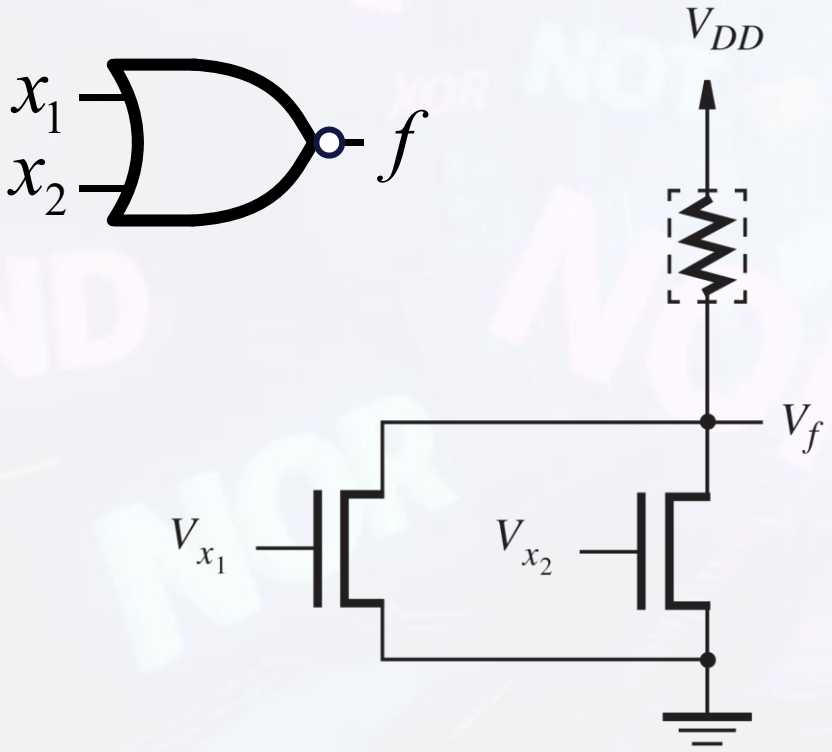
- **Transistors construct logic gates**
  - **NAND** and **AND** gates with “NMOS + PMOS”

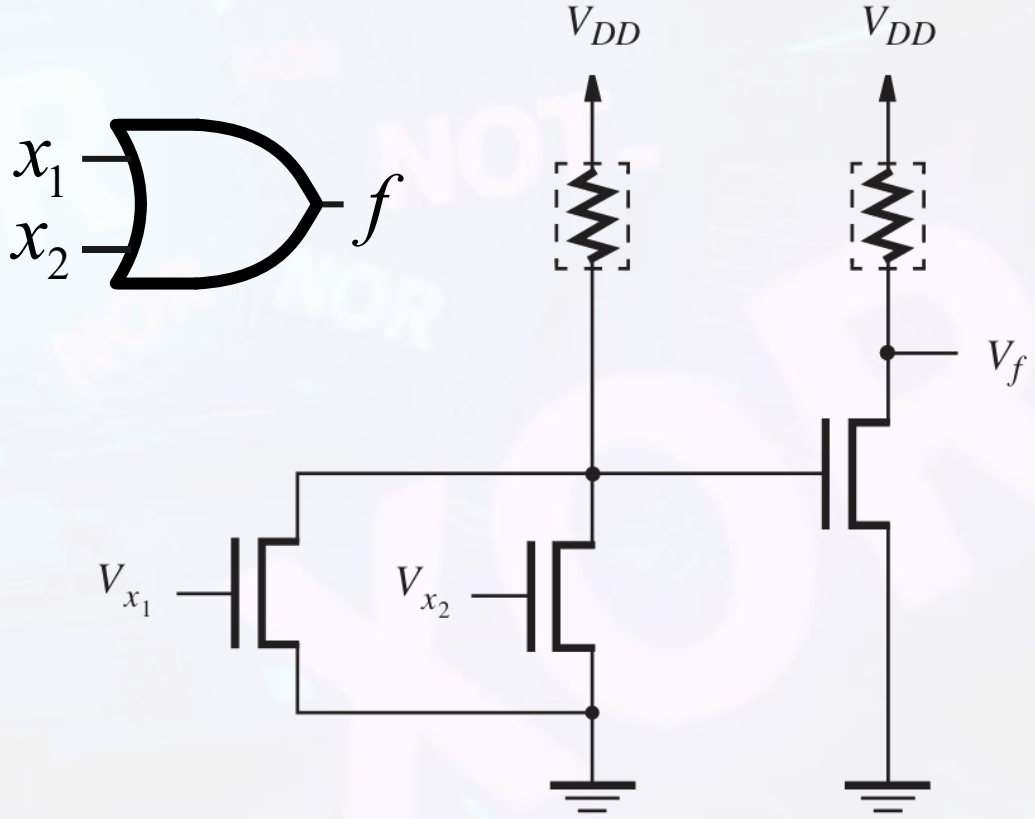




NAND and NOR Logic Networks

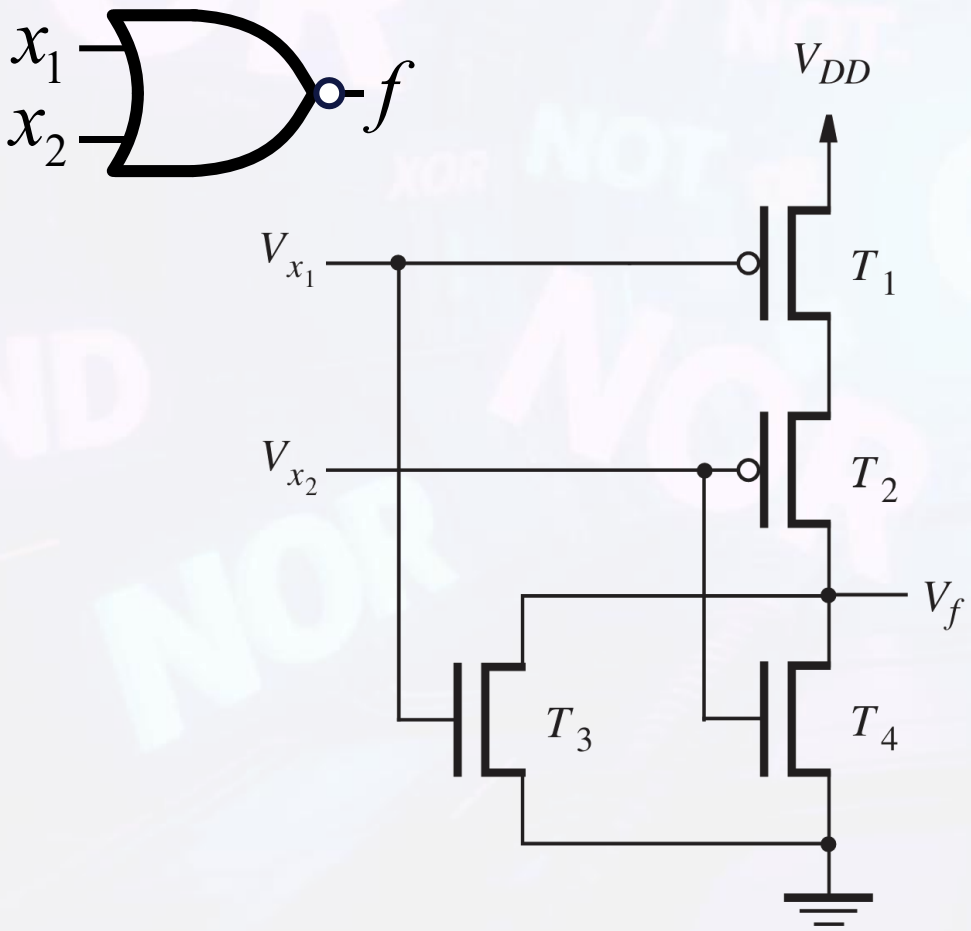
- **Transistors construct logic gates**
  - **NOR** and **OR** gates with NMOS only

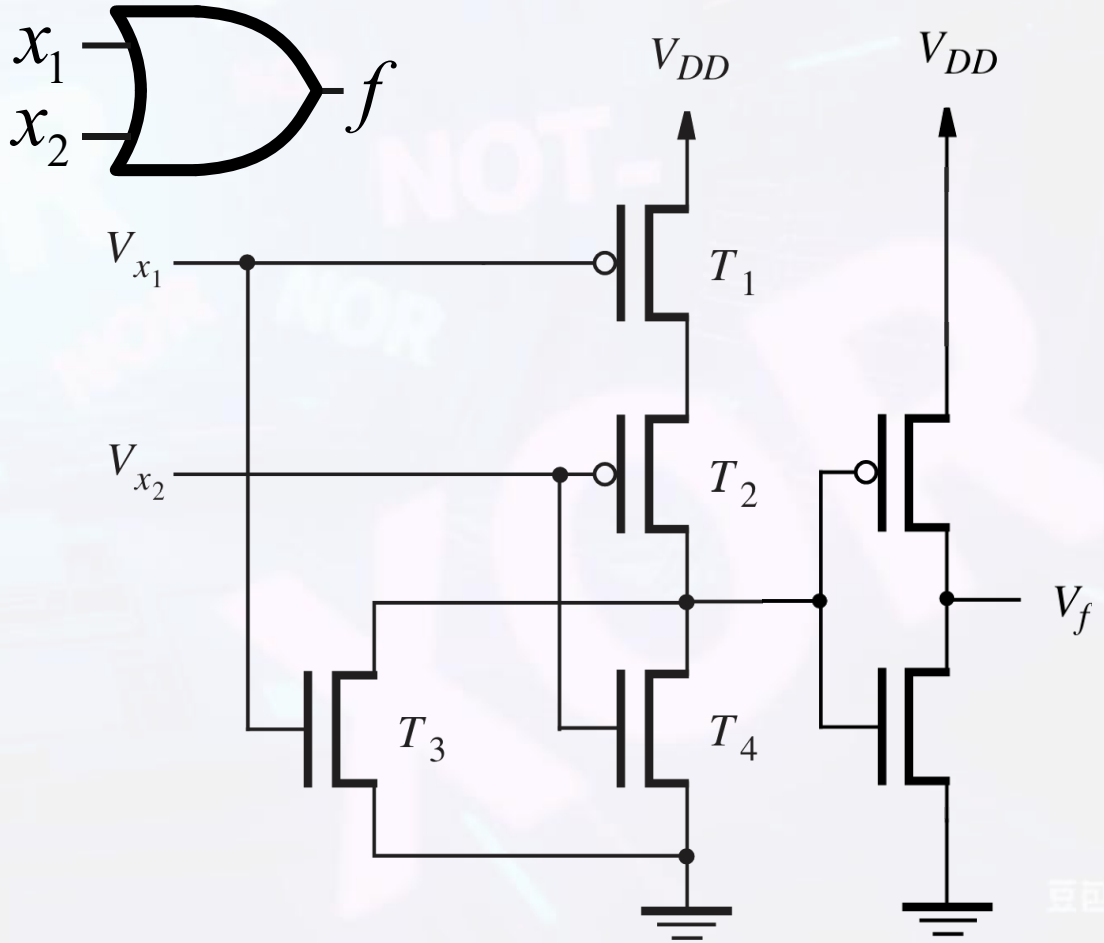




NAND and NOR Logic Networks

- **Transistors construct logic gates**
  - **NOR** and **OR** gates with “NMOS + PMOS”

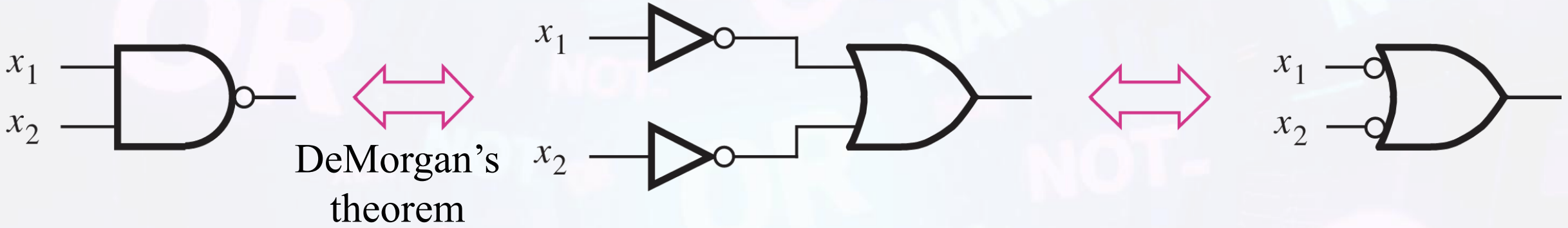




NAND and NOR Logic Networks

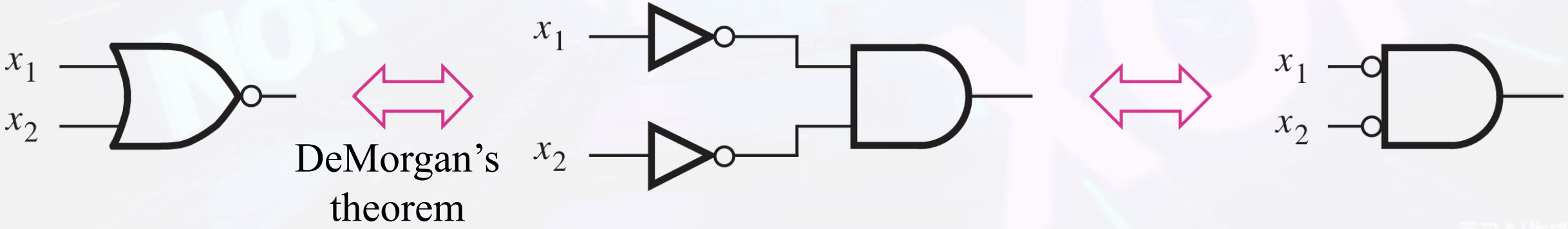
- **OR gate ↔ NAND gate**

$$\overline{x_1 x_2} = \bar{x}_1 + \bar{x}_2$$



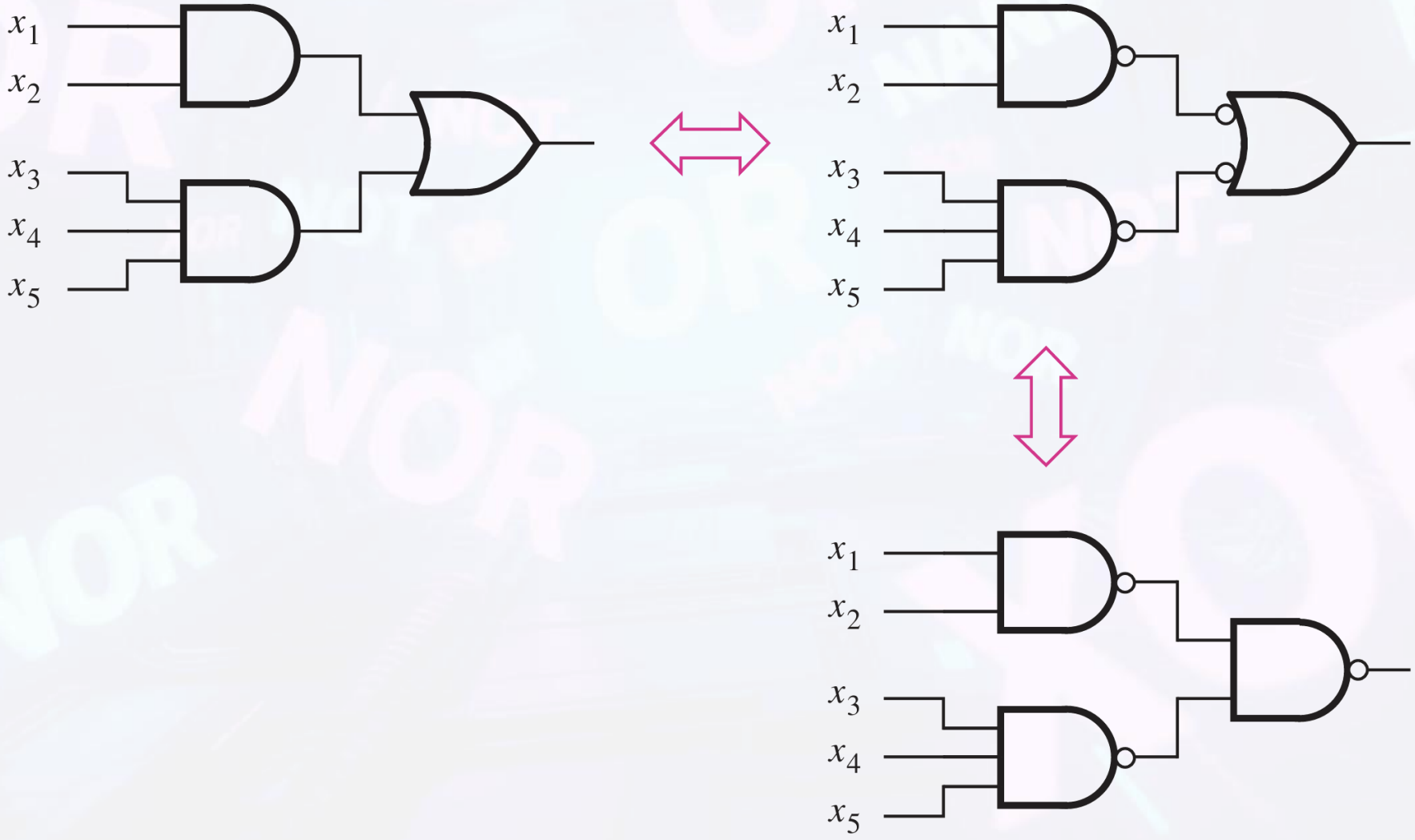
- **AND gate ↔ NOR gate**

$$\overline{x_1 + x_2} = \bar{x}_1 \cdot \bar{x}_2$$



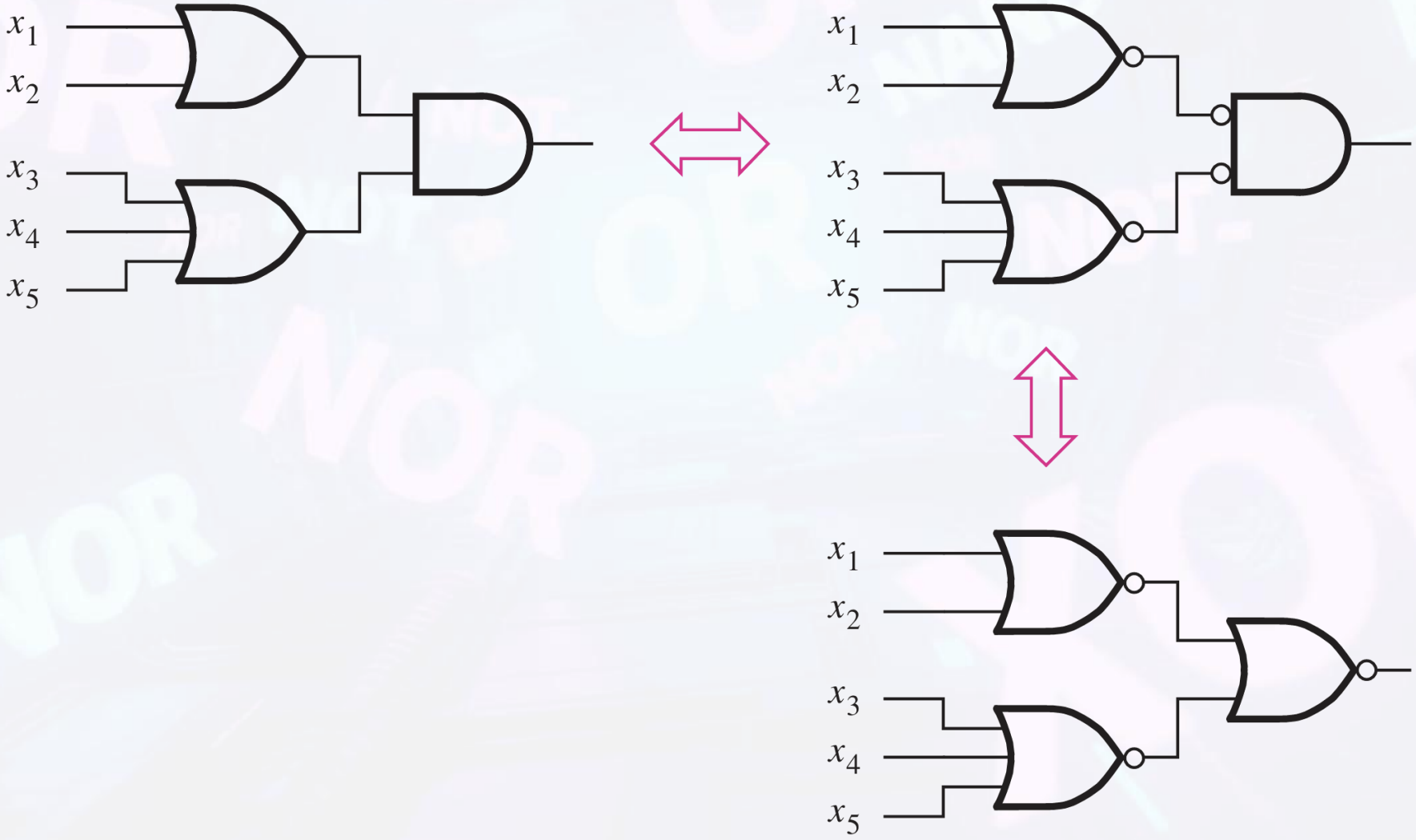
NAND and NOR Logic Networks

- **NAND gates implementing sum-of-products**



NAND and NOR Logic Networks

- **NOR gates implementing products-of-sum**



Practice:

Problem: Implement the function $f(x_1, x_2, x_3) = \sum m(2, 3, 4, 6, 7)$:

① using NAND gates only

② using NOR gates only

Solution:

Summary

- **Logic Variable**
  - Logic variable, function, operations
- **Representation of Logic Relations**
  - Truth table
  - Logic function
  - Logic gate network
  - Venn diagram
- **Conversion between Logic Representations**
  - Minterm & maxterm
  - Sum-of-product & product-of-sum
  - Cost of logic circuit
- **Boolean Algebra**
  - Axioms, theorems, properties
- **Minimization of Logic Expression**
  - Algebra manipulation
  - Karnaugh map
    - Minimize SOP
    - Minimize POS
    - Incompletely specified functions
- **Multiple-Output Circuits**
- **NAND and NOR Logic Networks**

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